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A physics-informed hybrid model for grinding force prediction

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ABSTRACT

Accurate prediction of grinding force is crucial for enhancing machining precision. Existing models often rely on simplified assumptions of rigid abrasive grains or mean chip thickness, neglecting the dynamic coupling between wheel surface randomness and grain wear. This paper proposes a physics-informed hybrid grinding force model (PHGFM). By integrating the geometric characteristics and wear behavior of grains, the wheel topography is reconstructed to accurately capture the evolution of the wheel surface topography during grinding. Furthermore, a novel undeformed chip thickness (UCT) model is proposed based on kinematic analysis and dynamic height distribution of grains. Moreover, the physical force computations are embedded into a physics-informed neural network (PINN), where a hybrid loss function incorporating both physical constraints and data-driven components is designed to achieve synergistic optimization of physics priors and data features. The proposed hybrid model demonstrates high predictive accuracy for grinding forces in brittle materials such as SiC ceramics under various grinding conditions. The average prediction errors for the normal and tangential grinding forces are 6.76 % and 7.28 %, respectively, while the minimum errors for the normal and tangential forces reach 3.35 % and 5.70 %, respectively.

1. Introduction

Advanced ceramic materials, particularly silicon carbide (SiC) ceramics, have drawn growing attention owing to their exceptional mechanical and physical properties [1,2]. These materials exhibit high strength, hardness and superior thermal stability, and have been widely applied in the aerospace field, automotive manufacturing, and electronics industries [3–5]. Grinding is usually the final stage in the advanced material manufacturing chain to attain superior surface quality and machining accuracy [6–8]. Nevertheless, the machining of SiC ceramics faces significant challenges due to its inherent brittleness [9,10] and the complex interaction between the grinding wheel and workpiece [11,12]. Among the many parameters influencing the grinding process, grinding force is a critical factor that directly affects machining performance, workpiece surface integrity, and wheel durability [13–16]. Therefore, accurately predicting the grinding force is essential for optimizing process parameters, minimizing tool wear, and enhancing grinding efficiency [17–19].

Owing to the stochastic characteristics of grains, early studies focused on empirical models that estimated grinding forces through regression between input variables and measured forces [20–22]. Subsequently, the pioneering empirical grinding force model proposed by Werner and Koenig [23] considered the effects of friction and chip formation and assessed the contribution of material hardness and chip formation properties to the normal and tangential forces. Malkin et al. [24] developed a ceramic grinding force model that combined plastic scribing and crack extension mechanisms, revealing the interaction between chip formation and surface damage. In all the above works, nevertheless, numerous experiments are usually needed to determine empirical constants, which greatly limit their applicability and require extensive experimental measurements and data regression for each new condition [25].

To address the inability of empirical models to capture the complex,

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Nomenclature

A_d, A_h	coefficient of the truncated normal distribution
A_s	surface coverage
$a_m(i, j)$	maximum undeformed chip thickness of grain (i, j)
a_p	grinding depth
$a_p(i, j)$	actual grinding depth of grain (i, j)
$a_t(i, j)$	instantaneous undeformed chip thickness of grain (i, j)
B	grinding wheel width
B_g	grinding width
C_l	length of the lateral cracks
C_h	depth of the lateral cracks
D	grinding wheel diameter
$d_g(i, j)$	size of abrasive grain (i, j)
d_{ga}	mean abrasive grain size
d_{gmax}	maximum abrasive grain size
d_{gmin}	minimum abrasive grain size
E	elastic modulus of the workpiece
E_g	elastic modulus of the grains
E^*	equivalent elastic modulus
$f_{en}(i, j)$	normal grinding force of grain (i, j) in elastic stage
$f_{et}(i, j)$	tangential grinding force of grain (i, j) in elastic stage
$f_{dn}(i, j)$	normal grinding force of grain (i, j) in ductile removal stage
$f_{dt}(i, j)$	tangential grinding force of grain (i, j) in ductile removal stage
$f_{bn}(i, j)$	normal grinding force of grain (i, j) in brittle removal stage
$f_{bt}(i, j)$	tangential grinding force of grain (i, j) in brittle removal stage
$F_{n,\ phy}$	physically predicted normal grinding force
$F_{t,\ phy}$	physically predicted tangential grinding force
$F_{n,\ data}$	experimentally measured normal grinding force
$F_{t,\ data}$	experimentally measured tangential grinding force
$\widehat{F}_n$	normal grinding forces predicted by the network
$\widehat{F}_t$	tangential grinding forces predicted by the network
h_c	z-coordinate of the grain center
$h_{ij}(0)$	initial protrusion heights of abrasive grain prior to wear
$h_{ij}(t)$	protrusion height of abrasive grain at wear time t
$h_{e\text{-}p}$	critical chip thickness of elastic-to-plastic transition
$h_{p\text{-}b}$	critical chip thickness of plastic-to-brittle transition
H	hardness of the workpiece material
H_g	hardness of the abrasive grain material
K_C	fracture toughness of the workpiece material
l_c	contact length between the grinding wheel and workpiece
l_e	trajectory length of the grain
M	wheel mesh size
N_x	number of active abrasive grains along the wheel circumferential direction
N_y	number of active abrasive grains along the wheel width direction
N_{eff}	total number of active abrasive grains in the contact area
N_{ds}	number of active grains per unit length along the contact arc
n_s	spindle speed
R_s	radius of the grinding wheel
$r(t)$	curvature radius at wear time t
S_p	spacing of adjacent grains
S_{pa}	mean inter-grain spacing
λ_1, λ_2	proportional coefficients
μ	interfacial friction coefficient
$u(t)$	mean decay factor of grain height distribution
v	relative sliding velocity
v_f	feed rate
v_s	wheel speed
$V_{ij}(0)$	initial volume of the abrasive grain
$\omega_{n,\ phy}$	weighting factor for physics-based constraints of normal grinding force
$\omega_{t,\ phy}$	weighting factor for physics-based constraints of tangential grinding force
$\omega_{n,\ data}$	weighting factor for data-driven loss of normal grinding force
$\omega_{t,\ data}$	weighting factor for data-driven loss of tangential grinding force
Z_{ij}	random offset vector
Z_b	z-coordinate of the grinding wheel binder surface
α	mean decay rate of grain height distribution
β	standard deviation decay rate of grain height distribution
σ_g, σ_h	standard deviation of grain size and initial grain protrusion height
$\sigma(t)$	standard deviation decay factor of grain height distribution
ξ_{I}, ξ_{II}	the proportions of type I and type II abrasive grains
$\delta(i, j)$	undeformed chip thickness produced by the height difference between the adjacent grains
$\delta_m(i, j)$	kinematic correction term for maximum undeformed chip thickness
$\delta_t(i, j)$	kinematic correction term for instantaneous undeformed chip thickness
υ, υ_g	Poisson's ratio of workpiece and grain
ϕ_i	initial phase angle of the abrasive grain
κ	empirical coefficient of curvature radius

multiscale grain-workpiece interactions and the microscopic material removal mechanisms, subsequent studies have increasingly turned to physics-based analytical models. This type of model achieves a deep elucidation of the force generation mechanisms by coupling the material constitutive behavior with the abrasive kinematics [26,27]. Rubenstein et al. [28] proposed that the total grinding force comprised both chip formation and frictional components. Building on this work, Li et al. [29] developed a grinding force model that incorporated both chip formation and frictional forces, and examined the influence of the undeformed chip thickness (UCT) parameter on the grinding force, which laid the foundation for the accurate prediction of the mechanism model. As this line of inquiry evolved, UCT was recognized as a critical factor affecting grinding force [30–32]. Agarwal et al. [33] established a UCT model based on the random height distribution of grains by considering the contact deformation, which laid the foundation for subsequent mechanical modeling. Li et al. [34] combined Hertzian contact theory with atomic force microscope (AFM) measurements to quantitatively determine the critical depths in SiC ceramics associated with the elastic-plastic and brittle-ductile transition. They then examined the influence of the UCT on the grinding stage and constructed a grinding force model from the material removal mechanism. UCT not only varies with grinding wheel kinematics, but also strongly influenced by the wheel's stochastic grain topography [35–37]. The wheel topography is typically characterized using statistical probability or distribution functions to describe the geometric parameters and spatial dispersion of abrasive grains [38,39]. Liu et al. [40] established a parametric assessment framework and three-dimensional modeling approach for grinding wheel topography by incorporating stochastic distributions of grain size, spatial location, orientation, and protrusion height. Innovatively, they proposed particle protrusion and particle protrusion area dispersion to characterize the distribution features of abrasive grains. Among these features, the protrusion height of grain serves as a key input parameter

in UCT modeling, and its precision underpins overall model accuracy [41–43]. To correct biases in classical UCT predictions, several authors have incorporated statistical distributions of grain height into UCT calculations. Agarwal et al. [44] developed a ceramic grinding UCT model incorporating the probabilistic distribution of grain protrusion heights and the elastic deformation analysis of the contact zone. Zhang et al. [45] further addressed the limitation of neglecting wheel morphology randomness by integrating wheel topography, workpiece discretization, kinematic analysis and single grain UCT calculation into a unified framework to obtain the complete UCT distribution. Nevertheless, during grinding, the height of grains changes dynamically due to wheel wear, leading to a continuous reduction of the effective grinding height of the grains, consequently influencing machining quality and grinding efficiency [46]. Due to the cumulative deviation in the UCT calculation, this leads to a cumulative amplification of the grinding force error. In belt grinding, Li et al. [47] introduced the gamma distribution to quantify the evolution of grain height wear. By incorporating flexible contact characteristics and kinematic analysis under non-uniform grain protrusion conditions, their approach provided valuable insights into UCT modeling. Huang et al. [48] measured the grinding wheel topography, and developed a force model that accounted for the non-uniform distribution of grain protrusion heights and energy perspective. This model provides a more effective theoretical support for the application of the axial ultrasonic vibration end grinding (AUVEG) technology in the processing of BK7 optical glass. Although these studies have advanced progress in modeling the stochasticity of wheel morphology and UCT, the coupled analysis of wheel stochastic morphology with abrasive wear during the grinding process has not yet been carried out, making it difficult for existing models to capture the real-time influence of grain height changes on UCT.

Furthermore, UCT is also an important criterion for distinguishing material removal mechanism and an increase in UCT leads to a shift in the removal mechanism from elastic deformation to brittle removal [49–51]. Zhang et al. [52] developed a grinding force model that incorporates both material-removal and plastic-stacking mechanisms, determined the critical depth of cut through scratch tests, and analyzed the mechanical behavior of single abrasive during the various material removal phases. Yang et al. [53] focused on zirconia (ZrO_2) ceramics in ultrasonic vibration assisted grinding (UVAG), modeling the grinding forces via single grain scratching tests to analyze the removal mechanisms in brittle materials. Zheng et al. [54] established a predictive model for the critical depth of the ductile-brittle transition (DBT) and developed a grinding force model that considered the stochastic distribution of grains as well as the interactions among multiple grains. These studies enhance the understanding of the DBT mechanism for machining brittle materials.

Although the physical analytical models have achieved certain results, the prediction accuracy is still facing a bottleneck [55,56]. On the one hand, most of the mechanism modeling methods are based on static or assumed unchanged wheel morphology and chip thickness, which are difficult to reflect the dynamic evolution of the abrasive grain change and grinding force state. On the other hand, with the complexity of the operating conditions, the difficulty of parameter calibration of the traditional physical model increases dramatically, and the ability of the generalization is limited. To address these shortcomings, Zhou et al. [57] proposed a genetic algorithm optimized back-propagation neural network (GA-BP), in which quantitative inputs such as grinding depth and qualitative inputs representing grinding direction were combined to predict grinding forces in creep-feed deep grinding. Fu et al. [58] analyzed the mechanical characteristics of ductile and brittle material removal stages and developed a dynamic grinding force model that accounts for abrasive trajectory and effective number of active grains. The model parameters were further optimized by genetic algorithm. In recent years, data-driven methods represented by deep learning have gained significant attention due to their powerful nonlinear mapping capabilities [59–61]. By autonomously mining the intrinsic correlations within massive amounts of process data, these methods have demonstrated distinct advantages in manufacturing process modeling [62,63]. Abyaneh et al. [64] analyzed the effect of different grinding wheel types, coolant and depth of cut on surface roughness and grinding force by a combination of machine learning algorithms (ML) such as gaussian process regression (GPR). The advantages of machine learning in predicting the grinding process of UNS S34700 steel with high accuracy were demonstrated. Prashanth et al. [65] addressed the prediction of grinding parameters for Inconel 51 under multiple cooling environments by employing ML techniques. Specifically, they combined experimental data-driven approaches with GPR to capture the nonlinear mapping between cooling conditions and grinding forces. Nevertheless, although data-driven approaches can accurately fit experimental data, they tend to rely on data in a black box manner, and the lack of physical interpretability and physical mechanisms still needs to be addressed [66–68]. Therefore, physics-informed neural network (PINN) and related artificial intelligence (AI) methods have been increasingly explored in the field of advanced manufacturing. He et al. [69] investigated the application of artificial intelligence in wafer manufacturing processes and discussed the future prospects of algorithms such as PINN. Lee et al. [70] studied the application of PINN in the grinding of brittle materials, developing deep learning models to predict surface roughness and grinding forces. Overall, incorporating physical constraints into data-driven models to enhance their generalization ability and physical interpretability has become an important development direction in advanced manufacturing process modeling.

This paper proposes a physics-informed hybrid grinding force model (PHGFM) to tackle the aforementioned challenges. To provide a comprehensive presentation of the research conducted, the rest of this paper is organized in the following manner. Section 2 first develops a coupled model that considers the coupling of grinding wheel topography and wear. Subsequently, the UCT calculation is refined from a kinematic standpoint by integrating motion trajectory equations derived from three-dimensional reconstruction of surface topography, thereby effectively compensating for biases introduced by abrasive grain wear. Next, a hybrid framework based on a PINN is developed, directly embedding the grinding force equations into the neural network. A dual attention mechanism is introduced to dynamically balance physical principles and data-driven learning, achieving a synergy between physical interpretability and data-driven benefits. Section 3 presents on SiC ceramics under multiple parameter combinations to validate the accuracy of PHGFM. Finally, Section 4 summarizes the major findings and presents the conclusions of this study.

2. Model formulation

To enable accurate prediction of grinding force, a comprehensive modeling framework is proposed, as illustrated in Fig. 1. In Section 2.1, the initial wheel topography is modeled to determine abrasive grain size, spacing and protrusion height (Section 2.1.1). Subsequently, a time-dependent reconstruction of the worn wheel surface is performed by incorporating the blunting curvature radius and the reduction in grain height, reflecting the progressive wear behavior of abrasive grains (Section 2.1.2). In Section 2.2, the UCT is refined through kinematic analysis that accounts for the coupling effect between abrasive wear and the stochastic distribution of grain topography. Finally, Section 2.3 calculates the grinding force for different material removal mechanisms and presents the PHGFM, which integrates physical priors with data-driven compensatory mechanisms to achieve high-precision grinding force prediction.

2.1. Reconstruction of random topography of grinding wheel considering wear passivation

This section develops a grinding wheel surface topography model that simultaneously considers the stochastic distribution of grains and

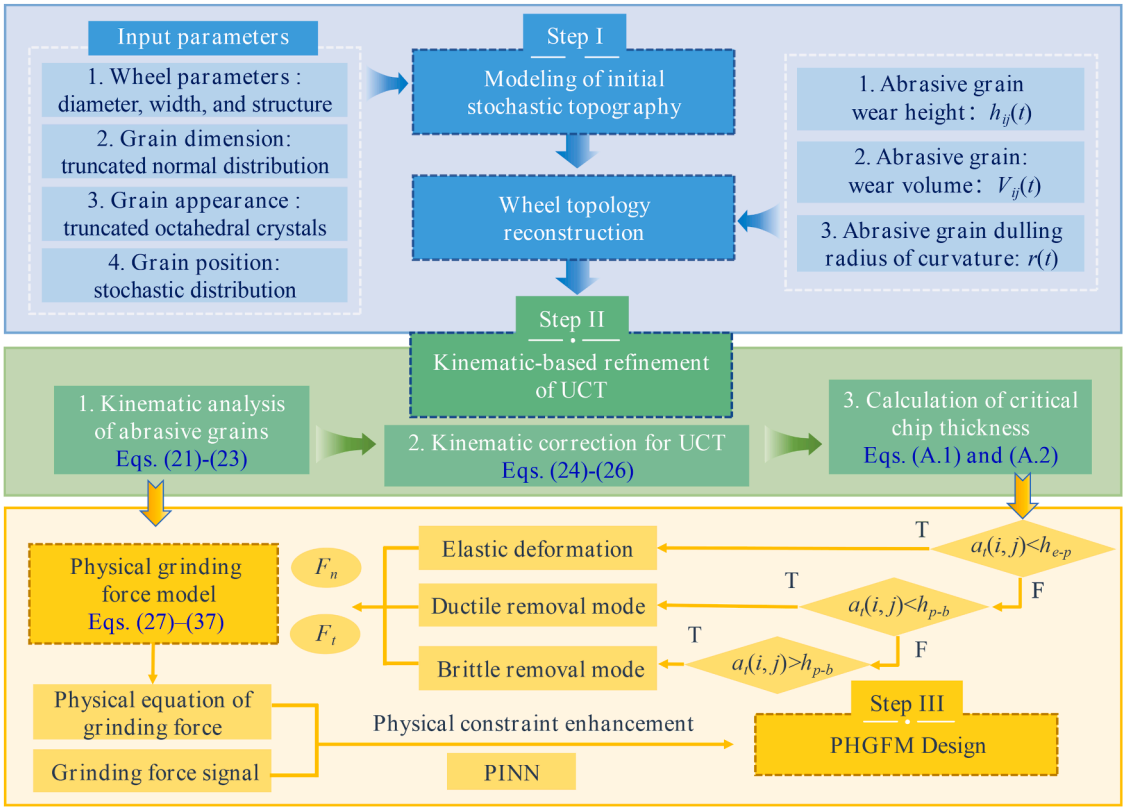


Fig. 1. Modeling framework for grinding force prediction. The modeling framework consists of three main steps: (1) establishing the wheel topography model to describe the stochastic distribution and wear evolution of abrasive grains, (2) refining the UCT based on kinematic analysis, (3) establishing PHGFM, which integrates physical modeling and data-driven learning for accurate force prediction.

their wear behavior. The initial surface geometry of the wheel is determined by defining its shape, size, grain spacing, and protrusion height. Subsequently, this model is coupled with time-dependent wear parameters to formulate a dynamic evolution model of the grinding wheel surface.

2.1.1. *Initial wheel topography*

The shape, size, protrusion height, location and grain distribution of the grinding wheel grains exhibit stochastic characteristics. These characteristics can significantly influence the grain movement trajectory and UCT [71–73], and are key factors affecting the grinding force. The grain shape of the diamond grinding wheel can be described as a truncated octahedral shape according to Liu et al. [40]. Those grains with a hexagonal top face are designated as type I grains, and those with a quadrilateral top face are as type II grains, as depicted in Fig. 2(a), which coincides with the results obtained from scanning confocal microscopy observations. The grinding wheel topography is observed by scanning confocal microscopy, as shown in Fig. 2(b). To accurately model the grinding wheel surface, the actual wheel surface is first discretized. The discretized modeling procedure for the wheel is elucidated in Fig. 2(c). The wheel's circular surface is extended into the circumferential and axial directions, and abrasive grains are assigned to each mesh element. The initial positions of the grains are denoted as (i, j), where i and j correspond to the circumferential and width directions, respectively.

Yang et al. [74] mentioned that the abrasive grain size d_g follows a truncated normal distribution, and its probability density function is given by:

$$f(d_g)=\begin{cases}\dfrac{A_d}{\sigma_g\sqrt{2\pi}}\exp\left[-\dfrac{1}{2}\left(\dfrac{d_g-d_{ga}}{\sigma_g}\right)^2\right], & d_{g\min}\le d_g\le d_{g\max},\\ 0, & \text{else},\end{cases} \tag{1}$$

where A_d is the coefficient of the truncated normal distribution. The maximum ($d_{g\max}$) and minimum ($d_{g\min}$) abrasive grain sizes can be obtained according to [75,76]. The mean abrasive grain size d_{ga} and standard deviation σ_g can be calculated by:

$$\begin{aligned} d_{ga}&=15200/M,\\ \sigma_g&=(d_{g\max}+d_{g\min})/6,\end{aligned} \tag{2}$$

where M is the wheel mesh size.

Grain spacing is an important factor in considering the relative position between grains. The variable S_p is modeled according to a normal distribution with a mean of S_{pa} and a standard deviation defined by (S_{pa} − d_{ga}). Specifically, the mean spacing of adjacent grains S_{pa} can be approximated as:

$$S_{pa}=\sqrt{\left[\xi_{\mathrm{I}}\left(\frac{3\sqrt{3}}{8}d_{ga}^2\right)+\xi_{\mathrm{II}}\left(\frac{1}{2}d_{ga}^2\right)\right]\Big/A_s}, \tag{3}$$

where ξ_{I} and ξ_{II} are the proportions of type I and type II abrasive grains, respectively. A_s is the surface coverage, defined as the ratio of the total projected area of the abrasive particles to the substrate's projected area.

Consequently, the total number of grains in the diamond tool can be calculated by:

$$N=\frac{\pi BD}{S_p^2}=\frac{\pi BDA_s}{\xi_{\mathrm{I}}\left(\frac{3\sqrt{3}}{8}d_g^2\right)+\xi_{\mathrm{II}}\left(\frac{1}{2}d_g^2\right)}, \tag{4}$$

where B and D denote the grinding wheel width and diameter, respectively.

The nonuniform distribution of abrasive grain protrusion height significantly affecting the grinding force. According to the study by Gao et al. [77], the initial protrusion heights of grains prior to wear follow a

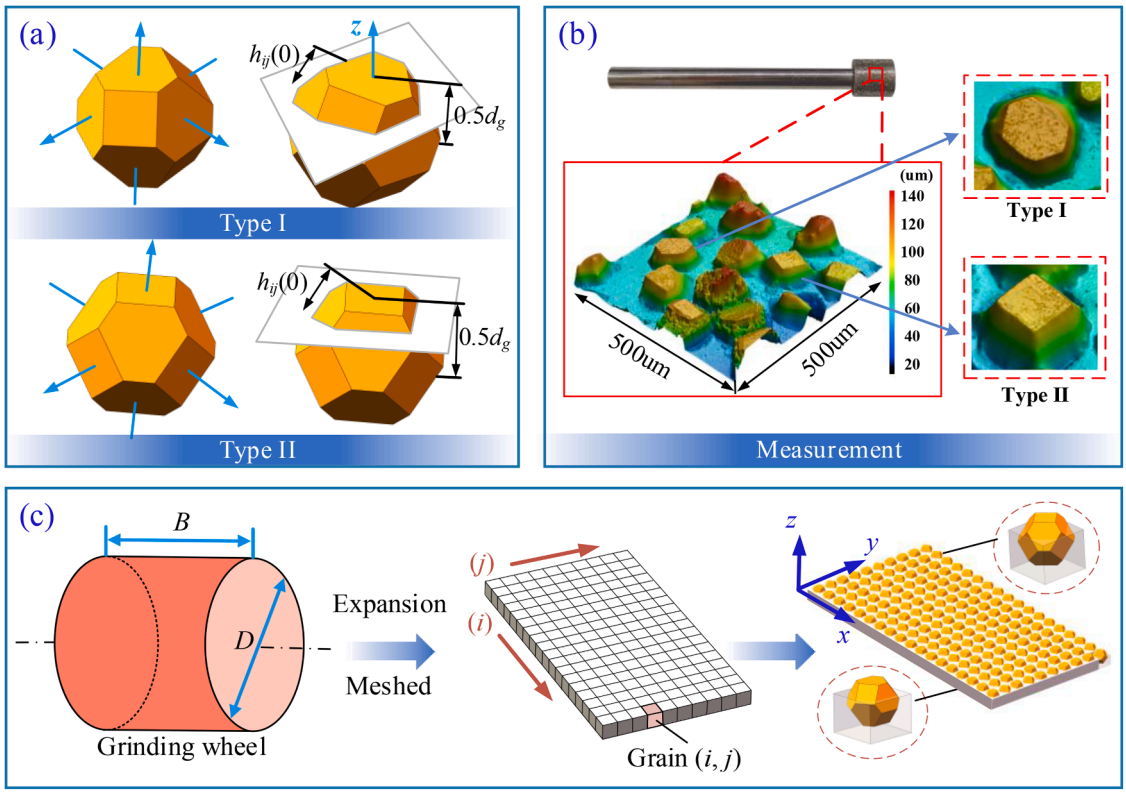


Fig. 2. Modeling of wheel topography. (a) Measuring abrasive grain size parameters by scanning confocal microscopy; (b)Schematic view of a representative abrasive grain; (c) Modeling process of the spatial grain distribution. The abrasive grain shapes can be classified into Type I grains with hexagonal top surfaces and Type II grains with quadrilateral top surfaces.

normal distribution, which can be expressed as:

$$h_{ij}(0) = d_g/2 + h_c - Z_b \tag{5}$$

A coordinate system is defined as illustrated in Fig. 2(c), in which h_c denotes the z-coordinate of the grain center. Z_b represents the z-coordinate of the grinding wheel binder surface and can be expressed as:

$$Z_b = d_{ga}\Big/3 - \underset{i,j}{max} Z_{ij}, \tag{6}$$

where Z_{ij} represents a spatial random vector that follows the Gaussian distribution $(0, ((S_{pa} - d_{ga})/3)^2)$.

Ultimately, the abrasive grain size, spacing, and protrusion height are measured via scanning confocal microscopy, and Gaussian fitting statistical analysis is performed to assess the initial wheel topography model accuracy. Detailed discussion is provided in Section 3.2.1.

The contact length l_c between the wheel and the workpiece directly influences the number of grains engaged in grinding, which can be expressed as [78]:

$$l_c = \left(1 + \frac{v_f}{\pi D n_s}\right)\sqrt{a_p D}, \tag{7}$$

where v_f is the feed rate, n_s represents the spindle speed, a_p is the grinding depth. Subsequently the actual number of grains engaged in grinding along the circumferential and width directions of the wheel, denoted as N_x and N_y, respectively, are given by:

$$(N_x, N_y)^{\mathrm{T}} = \frac{2}{d_g + S_p}(l_c, B_g)^{\mathrm{T}}, \tag{8}$$

where B_g is the grinding width. Ultimately, the total number of active grains is expressed as follows:

$$N_{eff} = N_x N_y = \frac{4B_g}{(d_g + S_p)^2}\left(1 + \frac{v_f}{\pi D n_s}\right)\sqrt{a_p D}. \tag{9}$$

2.1.2. Wheel topography reconstruction

In previous studies on grinding wheel topography modeling [39,40,82], abrasive grain parameters are typically treated as static, neglecting the blunting of grain tips due to wear and the consequent evolution of surface stochasticity. When the grinding parameters remain constant, abrasive wear becomes a critical factor contributing to the fluctuation of grinding force [79–81]. To account for the effect of grain wear, the wheel topography is reconstructed by incorporating the effects of grain wear.

As grinding progresses, abrasive grains experience relative motion with respect to the workpiece while enduring both normal and tangential forces. This interaction leads to gradual material removal at the grain tips, manifesting as a time-dependent reduction in the protrusion height. Prior to wear, the initial protrusion height $h_{ij}(0)$ of grain (i, j) can be expressed as:

$$h_{ij}(0) = \frac{A_h}{\sigma_h\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\frac{h_{ij}(0) - h_a}{\sigma_h}\right)^2\right], h_{\min} \le h_{ij}(0) \le h_{\max}, \tag{10}$$

where h_a and σ_h are the mean and standard deviation of the initial abrasive grain protrusion heights, A_h is the truncated factor of the normal distribution, and $h_{\min}$ and $h_{\max}$ denote the lower and upper truncation bounds, respectively. Nonetheless, this protrusion height model [40] does not account for the wear of the abrasive grain height of the wheel.

A clear description of the abrasive grain height reduction induced by wear is essential for accurately correcting the subsequent UCT calculations. Accordingly, the abrasive grain height wear decay factors $u(t) = u_0 - \alpha t$ and $\sigma(t) = \sigma_0 - \beta t$ are defined to describe the decay of the height

distribution, where u_0 and α represent the mean decay coefficients, and σ_0 and β represent the standard deviation decay coefficients. Accordingly, at any processing time t, the protrusion height $h_{ij}(t)$ is as follows:

$$h_{ij}(t) = \frac{A_h}{(\sigma_0 - \beta t)\sigma_h\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{(u_0 - \alpha t)h_{ij}(0) - h_a}{(\sigma_0 - \beta t)\sigma_h} \right)^2 \right]. \tag{11}$$

A schematic of a single grain is illustrated in Fig. 3(a), where the solid outline denotes its original shape. For the type I and type II abrasive grains that dominate the grinding wheel topography, their volume can be equivalent to a truncated pyramid with a height of $h_{ij}(t)$.

Assuming $r_g = 0.5d_g$, the initial volume of the abrasive grain (i, j) is given by:

$$V_{ij}(0) = \frac{h_{ij}(0)}{3}\left(A_1 + A_2 + \sqrt{A_1 A_2} \right), \tag{12}$$

where A_1 and A_2 denote the base and top cross-sectional areas of the frustum, respectively, which can be calculated by:

$$\begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \frac{3\sqrt{3}}{8} \begin{bmatrix} d_g^2 \\ \left[r_g - h_{ij}(0)\tan\theta \right]^2 \end{bmatrix}, \tag{13}$$

where θ is the half apex angle of grain.

In accordance with wear progresses, the originally sharp apex transitions into a rounded cap of larger curvature radius $r(t)$, approximating a spherical segment, as depicted in Fig. 3(b). During the grinding process, the curvature radius $r(t)$ directly influences the stress distribution and chip formation mechanism. Based on the modified Archard wear theory, the evolution of $r(t)$ with wear time t can be expressed as:

$$\frac{\mathrm{d}r(t)}{\mathrm{d}t} = \kappa \frac{pv}{H_g} \frac{1}{h_{ij}(t)}, \tag{14}$$

where κ is an empirical coefficient, p is the contact normal pressure, v is the relative sliding velocity, and H_g is the hardness of the grain material. The coefficient β is determined through grinding experiments. In surface grinding, the relative sliding velocity is $v = (v2\,f + v2\,s)^{1/2}$, where v_s is the wheel speed. The contact normal pressure p is defined as the grinding normal force F_n acting on the wheel, uniformly distributed over the contact area A_c. A_c can be estimated from the contact arc length l_c and the grinding width B_g. Therefore, p can be calculated by:

$$p = \frac{F_n}{B_g\left(1 + v_f/v_s\right)\sqrt{a_p D}}. \tag{15}$$

By substituting the mean protrusion height of the abrasive grains $\overline{h} = (\tau_0 - \alpha t)h_{ij}(0)$ into Eq. (14), and applying variable separation and integration under the initial condition $r(0) = r_0$, the curvature radius $r(t)$ can be expressed as:

$$r(t) = r_0 + \beta \frac{pv}{H} \frac{1}{h_{ij}(0)} \ln\left(\frac{\tau_0}{\tau_0 - at} \right). \tag{16}$$

Defining $\Delta r = r_g - r(t)$, the cross-sectional radius $r(z)$ at any intermediate height z can be expressed as:

$$r(z) = r_g - \Delta r \frac{z}{h_{ij}(t)}, z \in \left[0, h_{ij}(t)\right]. \tag{17}$$

Hence, the worn abrasive grain may be treated as a body of revolution, whose volume $V_{ij}(t)$ can be obtained as:

$$V_{ij}(t) = \int_0^{h_{ij}(t)} \pi \left(r_g - \Delta r \frac{z}{h_{ij}(t)} \right)^2 dz. \tag{18}$$

By incorporating the calculation of abrasive grain wear height, blunting curvature radius, and worn volume, the stochastic topography model of the grinding wheel is enriched, providing a more realistic representation of its morphological evolution during the grinding process. To provide an intuitive visualization of the dynamic reconstruction of the wheel topography, the three-dimensional wheel topography is numerically simulated. To ensure visual clarity while capturing microscopic surface features, a representative sampling band with a width of 1 mm is selected. The initial topography is shown in Fig. 3(c), whereas the topography after wear is presented in Fig. 3(d). A color gradient is employed to represent the protrusion height of grains. As the grinding process progresses, the protrusion height exhibits a pronounced

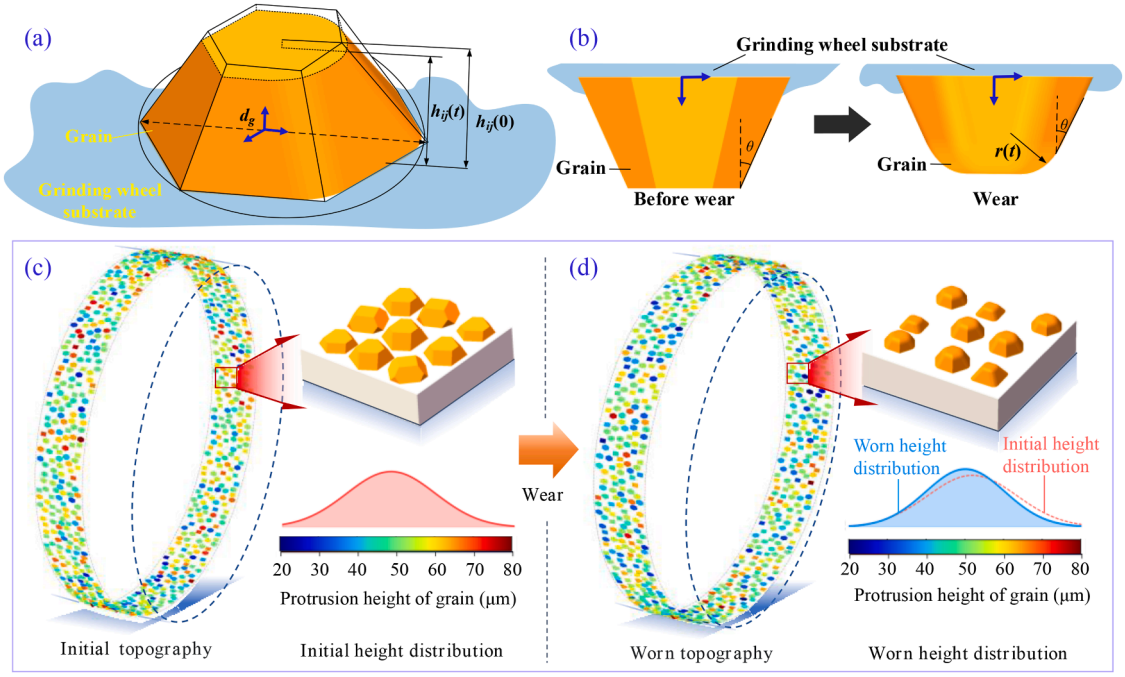


Fig. 3. Schematic of the abrasive grain model for wheel topography modeling. (a) Geometric model of a single grain; (b) Evolution of grain geometry during wear; (c) Initial topography of the grinding wheel; (d) Surface topography after wear. The wear simulation captures the dynamic evolution of the wheel surface by accounting for both the gradual blunting of grain tips and protrusion height attenuation during the grinding process.

decreasing trend, manifested by a gradual transition in color from red-dominated high peaks to blue-green tones corresponding to lower elevations. Concurrently, the geometric profile of grain tips evolves from initially sharp asperities to blunted, dome-like structures. This evolution process provides an intuitive visualization of the height decay and tip-blunting mechanisms established in this section, illustrating the topography evolution of the grinding wheel surface during the grinding process.

2.2. Kinematic-based refinement of UCT

This section develops an improved UCT refinement method based on a detailed analysis of abrasive grain kinematics. In the classical UCT model [33,45], UCT is typically calculated under the assumption that the protrusion height of each abrasive grain remains constant. This assumption neglects the combined effects of grain wear and randomness of grinding wheel topography, which may cause cumulative errors in force prediction. In this section, a novel UCT calculation strategy is proposed, which integrates time-dependent decay of grain protrusion height with grain motion analysis to dynamically correct UCT computation errors. This method enables more accurate estimation of the UCT.

Previous studies have accounted for the influence of grain protrusion height, with the UCT calculated as follows [41]:

$$a_m(i,j) = 2S_p\frac{v_f}{v_s}\sqrt{\frac{a_p(i-1,j)}{D}} - \delta(i,j), \tag{19}$$

$$\delta(i,j) = a_p(i-1,j) - a_p(i,j), \tag{20}$$

where $a_m(i, j)$ represents the maximum UCT of the grain (i, j), and $\delta(i, j)$ is the UCT produced by the height difference between the adjacent grains. $a_p(i-1, j)$ and $a_p(i, j)$ denote the actual grinding depths of grains $(i-1, j)$ and (i, j), respectively.

However, owing to the inherent randomness of grain protrusion height and the influence of grinding kinematics, the actual path lengths that adjacent grains travel during grinding differ. Moreover, the progressive wear of grain height over time further affects the accuracy of UCT calculations. To remedy these constraints, a more precise UCT prediction model is developed by incorporating both grinding kinematics and time-dependent abrasive wear.

During surface grinding, multiple abrasive grains participate in material removal. To account for the kinematic interference between grains, a global system fixed on the workpiece is established to describe the motion trajectory of grains, as depicted in Fig. 4. The origin O_g corresponds to the location at which the abrasive grain initially engages with the workpiece surface. X_g-axis indicates the feed direction, Y_g-axis corresponds to the wheel width direction, and Z_g-axis is perpendicular to the workpiece surface.

Considering the joint effects of the wheel feed and rotational motions, and incorporating the random heights of grains, the trajectory of the grain tip can be described as:

$$\begin{bmatrix} x(i,j) \\ y(i,j) \\ z(i,j) \end{bmatrix} = \begin{bmatrix} v_f t + [R_s + h_{ij}(t)]\sin(2\pi n_s t + \phi_{ij}) \\ \text{constant} \\ [R_s + h_{ij}(t) - a_p] - [R_s + h_{ij}(t)]\cos(2\pi n_s t + \phi_{ij}) \end{bmatrix}, \tag{21}$$

where R_s is the radius of the grinding wheel and ϕ_{ij} is the initial phase angle of the abrasive grain.

Accordingly, the velocity equation of the grain (i, j) is as follows:

$$\begin{bmatrix} v_x(i,j) \\ v_y(i,j) \\ v_z(i,j) \end{bmatrix} = \begin{bmatrix} v_f + [R_s + h_{ij}(t)]\cos(2\pi n_s t + \phi_{ij}) + \dot{h}_{ij}(t)\sin(2\pi n_s t + \phi_{ij}) \\ 0 \\ \dot{h}_{ij}(t)[1 - \cos(2\pi n_s t + \phi_{ij})] - [R_s + h_i(t)]\sin(2\pi n_s t + \phi_{ij}) \end{bmatrix}. \tag{22}$$

Considering the influence of the motion trajectory on the UCT, a kinematic correction term $\delta_m(i, j)$ is introduced to represent the maximum UCT at the same position between grain $(i-1, j)$ and grain (i, j). $\delta_m(i, j)$ is given by:

$$\delta_m(i,j) = [x_g(i-1,j) - x_g(i,j)]\tan\varepsilon_{\max}, \tag{23}$$

where $x_g(i-1, j)$ and $x_g(i, j)$ denote the displacements of grain $(i-1, j)$ and grain (i, j), respectively. $\tan\varepsilon_{\max}$ represents the maximum tangential slope of the trajectory. The tangential slope of the trajectory can be described by the velocity vector as follows: $\tan\varepsilon = v_z(i, j)/v_x(i, j)$.

By considering the overlap of grain trajectories and the effect of grain height fluctuations on the UCT during the instantaneous grinding process, the instantaneous kinematic correction term $\delta_t(i, j)$ can be finally expressed as:

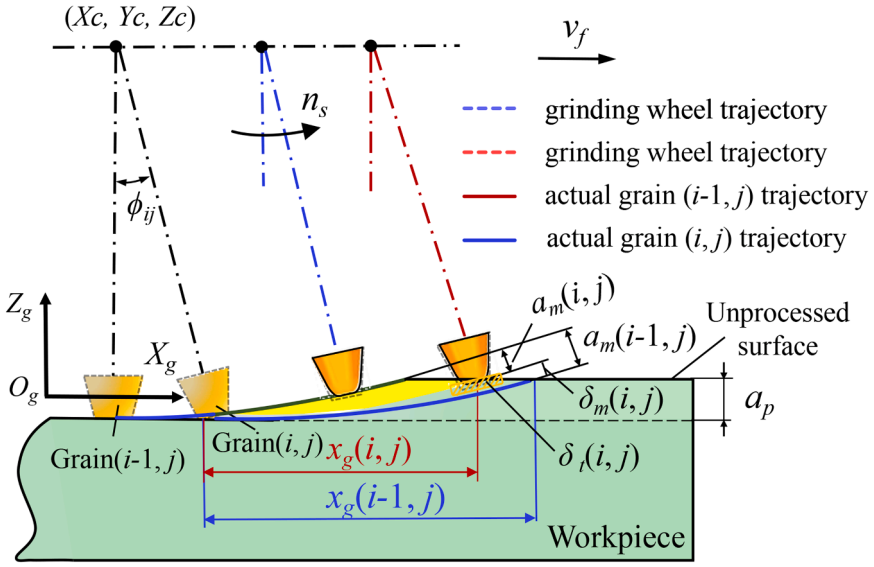


Fig. 4. Schematic of the superimposed trajectories of multiple abrasive grains, where (X_c, Y_c, Z_c) is the center coordinate of the grinding wheel). Differences in protrusion height and phase angle among adjacent abrasive grains lead to variations in their contact paths, which form the basis for calculating the motion correction term $\delta_m(i, j)$.

$$\delta_t(i,j) = \left(1+\frac{v_f}{v_s}\right)\left[\sqrt{2a_p(i-1,j)\cdot\left(R_s+h_{i-1j}(t)\right)} - \sqrt{2a_p(i,j)\cdot\left(R_s+h_{ij}(t)\right)}\right]\frac{v_z(i,j)}{v_x(i,j)}. \tag{24}$$

Different grain heights result in varying instantaneous grinding depths, as illustrated in Fig. 5(a). Accordingly, the actual grinding depth $a_p(i, j)$ of grain (i, j) can be expressed as:

$$a_p(i,j) = a_p - \left[h_{ij}(t)_{\max} - h_{ij}(t)\right], \tag{25}$$

where $h_{ij}(t)_{\max}$ denotes the maximum protrusion height.

Consequently, the final expression for the corrected instantaneous UCT $a_t(i, j)$ is given by:

$$a_t(i,j) = 2S_p\frac{v_f}{v_s}\sqrt{\frac{a_p(i-1,j)}{D}} - \delta_t(i,j). \tag{26}$$

Throughout the grinding path of grain (i, j), the dynamic variation of the instantaneous UCT leads to transitions between material removal mechanisms. As illustrated in Fig. 5(b), the material removal mechanisms of the grain (i, j) can be classified into: (i) elastic deformation, (ii) ductile removal, and (iii) brittle removal. These mechanisms are distinguished based on the instantaneous UCT in relation to the corresponding critical depth.

By evaluating the relationship between $a_t(i, j)$ and the critical depths ($h_{e\text{-}p}$, $h_{d\text{-}b}$), the different material removal mechanisms of the grain (i, j) can be determined. The detailed calculations of the critical depths ($h_{e\text{-}p}$, $h_{d\text{-}b}$) are provided in Appendix A.

(i) For $a_t(i, j) < h_{e\text{-}d}$, grain (i, j) experiences elastic deformation stage.
(ii) For $h_{e\text{-}d} < a_t(i, j) < h_{d\text{-}b}$, grain (i, j) performs in ductile removal mode.
(iii) For $a_t(i, j) > h_{d\text{-}b}$, grain (i, j) performs in brittle removal mode.

2.3. PHGFM design

This section proposes a PHGFM that integrates physical mechanism modeling with data-driven learning to achieve both physical interpretability and high prediction accuracy. Grinding force mechanism modeling methods are predominantly based on the assumption of invariant wheel topography and chip thickness [13,48], which restricts their capability to reflect the dynamic changes in wheel topography and grinding force throughout the process. To improve prediction accuracy, a novel PHGFM is proposed, which integrates physical interpretability with the powerful capabilities of deep learning in modeling complex nonlinear relationships. Specifically, the model first formulates grinding force equations for elastic, plastic, and brittle stages, and then obtains the total grinding force by summing the contributions of all active grains within the contact zone. The physical computation model is subsequently embedded into a PINN framework, in which a hybrid loss function is designed to balance physical constraints and data-driven learning. Moreover, a dual attention mechanism is introduced to dynamically balance the fusion of physical and data-driven information, enabling the network to adaptively focus on key features. As a result, the proposed PHGFM achieves both physical interpretability and the advantages of data-driven modeling.

The physical grinding force model is formulated corresponding to different material removal mechanisms. During the elastic deformation stage, the workpiece surface elastically deforms due to the frictional interaction with the grain. The normal grinding force $f_{en}(i, j)$ exerted by a single grain is given by [54]:

$$f_{en}(i,j) = \frac{2}{3}E^*\left[2d_g(i,j)a_t^3(i,j)\right]^{1/2}, \tag{27}$$

where $d_g(i, j)$ denotes the size of abrasive grain (i, j). Accordingly, the tangential grinding force $f_{et}(i, j)$ is expressed as follows:

$$f_{et}(i,j) = \frac{2}{3}\mu_e\frac{EE_g}{E\left(1-\nu_g^2\right)+E_g\left(1-\nu^2\right)}\left[2d_g(i,j)\cdot a_t^3(i,j)\right]^{1/2}, \tag{28}$$

where μ_e represents the interfacial friction coefficient.

In the ductile removal mode, material removal is achieved through compressive stress-induced plastic flow. The normal grinding force f_{dn} and the tangential grinding force f_{dt} generated by a single grain are expressed as:

$$f_{dn}(i,j) = \frac{3\pi}{4}\left(\frac{H^4}{E}\right)^{1/3}\left[2r_a a_t(i,j) - a_t^2(i,j)\right], \tag{29}$$

$$f_{dt}(i,j) = u_d\frac{3\pi}{4}\left(\frac{H^4}{E}\right)^{1/3}\left[2r_a a_t(i,j) - a_t^2(i,j)\right]. \tag{30}$$

where μ_d represents the interfacial friction coefficient during the ductile removal mode.

In the brittle removal mode, high grinding loads lead to the initiation of brittle fracture and crack propagation, as illustrated in Fig. 5(c), where the critical crack propagation force is considered as the normal grinding force. The length and depth of the lateral crack, denoted as C_l

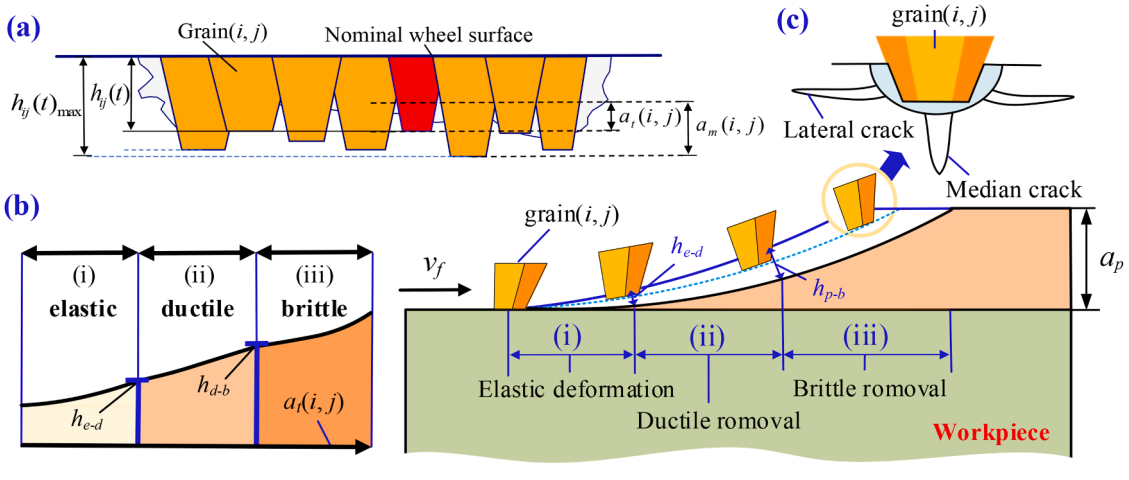


Fig. 5. Schematic illustration of grain material removal mechanisms: (a) Grain protrusion height; (b) Removal mechanisms that an individual grain may experience; (c) Schematic diagram of brittle fracture. By comparing the critical depth with the instantaneous UCT, the material removal mechanism of the grain can be determined.

and C_h, can be calculated by [53]:

$$C_l = \lambda_1 \left(\frac{f_{bn}}{K_C}\right)^{\frac{3}{4}}, C_h = \lambda_2 \left(\frac{f_{bn}}{H}\right)^{\frac{1}{2}}, \tag{31}$$

where λ_1 and λ_2 are proportional coefficients associated with the material properties, and f_{bn} represents the normal grinding force exerted by a single grain in the brittle removal mode.

Based on the material removal rate model, the macroscopic material removal per unit time is considered equal to the microscopic material removal, as follows:

$$a_p B_g v_f = 2C_l C_h l_e N_{ds} B_g v_s, \tag{32}$$

where N_{ds} is the number of active grains per unit length along the contact arc, and l_e is the trajectory length of the grain. It can be calculated for the corresponding grinding time t as:

$$l_e = \int_0^t \sqrt{(v_x(t))^2 + (v_z(t))^2}\, dt\,. \tag{33}$$

Accordingly, the normal grinding force f_{bn} and the tangential grinding force f_{bt} generated by a single grain during brittle removal can be expressed as:

$$f_{bn} = K_c^{3/5} H^{2/5} \left(\frac{a_p B_g v_f}{2\lambda_1 \lambda_2 l_e}\right)^{4/5}, \tag{34}$$

$$f_{bt}(i,j) = u_b K_c^{3/5} H^{2/5} \left(\frac{a_p B_g v_f}{2\lambda_1 \lambda_2 l_e}\right)^{4/5}. \tag{35}$$

where μ_b represents the interfacial friction coefficient in the brittle removal mode.

As a grain engages with and disengages from the workpiece, its spatial trajectory governs the continuous temporal evolution of the $a_t(i, j)$. This evolution directly induces fluctuations in the grinding force magnitude. Moreover, as $a_t(i, j)$ traverses the critical depths, individual grains undergo dynamic transitions among elastic deformation, ductile material removal, and brittle fracture regimes. Consequently, both the force generation mechanisms and the corresponding analytical formulations for the single grain grinding forces force shift in real time. To evaluate the total grinding force at an arbitrary time t, based on the correlation between the instantaneous UCT $a_t(i, j)$ and the critical depths ($h_{e\text{-}p}$, $h_{d\text{-}b}$), the grains are classified into three subsets: G_e, G_p, and G_b, corresponding to the grains in the elastic deformation stage, ductile removal mode, and brittle removal mode, respectively. By categorizing all active grains involved in grinding and summing their contributions, the physically predicted normal grinding force $F_{n,\,phy}$ can be calculated as:

$$\begin{aligned} F_{n,phy} = & \sum_{(i,j)\in G_e} \frac{2}{3} E^* \left[2d_g(i,j)\cdot a_t^3(i,j)\right]^{1/2} \\ & + \sum_{(i,j)\in G_p} \frac{3\pi}{4}\left(\frac{H^4}{E}\right)^{1/3} \left[2r_a a_t(i,j) - a_t^2(i,j)\right] \\ & + \sum_{(i,j)\in G_b} K_c^{3/5} H^{2/5} \left(\frac{a_p B_g v_f}{2\lambda_1 \lambda_2 l_e}\right)^{4/5}. \end{aligned} \tag{36}$$

Accordingly, the physically predicted tangential grinding force $F_{t,\,phy}$ is given by:

$$F_{n,phy} = \sum_{(i,j)\in G_e} f_{et}(i,j) + \sum_{(i,j)\in G_p} f_{pt}(i,j) + \sum_{(i,j)\in G_b} f_{bt}(i,j). \tag{37}$$

PINN integrates knowledge from physical models with neural networks to capture complex system behaviors and nonlinearities [83–85]. To achieve high-precision prediction of grinding forces while preserving the model's physical interpretability, the grinding force physical model (Eqs. (38) and (39)) is embedded within the PINN framework. By jointly constraining the governing physical laws and experimental observations, a hybrid loss function that fuses physics-based priors with data-driven terms is constructed. This enables the simultaneous optimization of prediction accuracy and physical interpretability, even with limited experimental data.

Within the proposed PHGFM framework, the PINN network input comprises the observable grinding parameter vector $\boldsymbol{X}_p = [v_f, a_p, v_s]$, the extractable grinding wheel morphology feature vector $\boldsymbol{X}_t = [d_g, S_p, h_{ij}(0), h_{ij}(t), N_{eff}, r(t), V_{ij}(t)]$, and the material parameter vector $\boldsymbol{X}_m = [E, H, K_C, \mu_e, \mu_d, \mu_b]$, which together form the complete input vector $\boldsymbol{X} = [\boldsymbol{X}_p, \boldsymbol{X}_t, \boldsymbol{X}_m]$. In conventional PINN architectures, multilayer perceptron (MLP) processes all input features with fixed weights, lacking mechanisms to selectively emphasize critical input features [85,86]. In this study, a dual attention mechanism is introduced within the network design. On the one hand, an attention mechanism is introduced between input and hidden layers, enabling the model to focus on the most critical input features and thereby enhancing the influence of key information during network propagation. The core idea of the feature attention mechanism is to dynamically adjust the weight of each input feature according to its contribution to the loss function, thereby enabling adaptive learning and dynamic optimization of the input sequence weights [87–89]. During training, the input feature vector $\boldsymbol{X}$ is first normalized to improve the stability of attention weight learning. The normalized vector $\boldsymbol{X}_n$ is then fed into a MLP to extract importance scores, which are subsequently transformed into a trainable feature attention weight vector $\boldsymbol{\alpha}$ via the softmax function. Eventually, the weights are applied to the input features through an element-wise product to obtain the weighted features $\widetilde{\boldsymbol{X}}$, which can be described as:

$$\widetilde{\boldsymbol{X}} = \boldsymbol{\alpha} \odot \boldsymbol{X}, \tag{38}$$

where $\odot$ denotes element-wise multiplication. The feature weights are automatically adjusted via backpropagation to minimize the loss while maximizing the weights, thereby enabling the network to emphasize critical physical parameters.

On the other hand, an attention mechanism is incorporated into the loss function design by introducing trainable weighting parameters. These parameters are automatically learned by the model to dynamically balance the relative contributions of the data-driven loss and the physics-based constraint loss, thereby achieving a trade-off between physical interpretability and predictive accuracy. It is worth noting that the physical constraint term employed in this study differs from the conventional partial differential equation residual form commonly used in classical PINN. In this study, the grinding force physical model is directly embedded into the loss function as prior knowledge to constrain the predicted outputs of the neural network. Specifically, separate physical constraint losses and data supervision losses are introduced for the predicted normal and tangential grinding forces components, forming a decoupled hybrid loss function.

The physical loss terms for the normal and tangential forces are defined as:

$$\mathscr{L}_{n,phy} = \| \widehat{F}_n - F_{n,phy} \|^2, \ \mathscr{L}_{t,phy} = \| \widehat{F}_t - F_{t,phy} \|^2, \tag{39}$$

where $\widehat{F}_n$,$\widehat{F}_t$ represent the normal and tangential grinding forces predicted by the network, respectively.

The data loss terms for the normal and tangential forces are given by:

$$\mathscr{L}_{n,data} = \| \widehat{F}_n - F_{n,mea} \|^2, \ \mathscr{L}_{t,data} = \| \widehat{F}_t - F_{t,mea} \|^2, \tag{40}$$

where $F_{n,\,mea}$ and $F_{t,\,mea}$ are the experimentally measured normal and tangential grinding forces.

Weight factors $\omega_{n,\,data}$, $\omega_{t,\,data}$, $\omega_{n,\,phy}$, and $\omega_{t,\,phy}$ are introduced to balance the contributions of the physical model and data-driven components. Accordingly, the total loss function is formulated as:

$$\mathscr{L}_{total} = \omega_{n,phy}\mathscr{L}_{n,phy} + \omega_{n,data}\mathscr{L}_{n,data} + \omega_{t,phy}\mathscr{L}_{t,phy} + \omega_{t,data}\mathscr{L}_{t,data}, \quad (41)$$

where $\omega_{n,\,phy}$ and $\omega_{t,\,phy}$ represent the weighting factors for the physics-based constraints of the normal and tangential grinding forces, respectively. These factors regulate the influence of physical priors in the total loss, ensuring that network predictions to maintain both high accuracy and adherence to the physical laws governing the grinding force model. Similarly, $\omega_{n,\,data}$ and $\omega_{t,\,data}$ represent the weighting factors for the data-driven losses of the normal and tangential grinding forces, controlling the contribution of experimental observations during training to fully exploit measurement information and enhance prediction accuracy. All four weighting coefficients are automatically learned through a loss weight attention mechanism and dynamically adjusted during training, enabling an organic integration of physical interpretability with the advantages of data-driven modeling. Specifically, the four trainable parameters are assembled into a vector $\boldsymbol{s} = [\omega_{n,\,data}, \omega_{t,\,data}, \omega_{n,\,phy}, \omega_{t,\,phy}]$, and passed through a softmax function to obtain the final weight vector, as follows:

$$\omega = \left[\omega_{n,data}, \omega_{t,data}, \omega_{n,phy}, \omega_{t,phy}\right] = softmax(\boldsymbol{s}). \quad (42)$$

To further provide an intuitive characterization of the overall contribution evolution of physics-based constraints and data-driven learning during the training process, the weight values in the loss function are divided into an aggregated physics-based constraint weight ω_{phy} and an aggregated data-driven weight ω_{data}. Specifically, they are defined as $\omega_{phy} = \omega_{n,\,phy} + \omega_{t,\,phy}$, and $\omega_{data} = \omega_{n,\,data} + \omega_{t,\,data}$. These weights can dynamically reflect the emphasis of the hybrid model during training process, thereby providing a basis for analyzing the dynamic balance between physics-based constraint and data-driven learning during the model optimization process. The weights vector $\boldsymbol{\omega}$ is updated via backpropagation, which not only minimizes the overall loss but also automatically enhances the contribution of the corresponding loss terms. This enables a dynamic balance between physical constraints and data fitting, ultimately achieving a synergistic optimization of predictive accuracy and physical consistency. Fig. 6 illustrates the overall architecture of PHGFM, which includes the input features, dual attention module, neural network predictions, physics-based and data-driven loss functions, and optimization process.

3. Experimental setup and validation

In this section, the superiority of the proposed theoretical model is verified through a series of experiments. An experimental platform is first constructed, with comprehensive details of the workpiece material, grinding wheel specifications, machining conditions machining conditions, as well as the scale of the experimental dataset used for model training and its partitioning strategy provided (Section 3.1). Subsequently, the grinding wheel topography model is validated through comparative analysis and statistical fitting of the experimental measurements (Section 3.2). Ultimately, the predictive capability of the proposed PHGFM is experimentally verified by comparing its predictions with the measured grinding forces, and is further benchmarked against existing theoretical grinding force models (Section 3.3). This section presents the overall experimental design and validation procedure.

3.1. Experimental equipment and conditions

This section describes the experimental equipment, workpiece materials, wheel specifications, and grinding parameters used for experimental validation. Detailed information is provided regarding the machine tools, measurement instruments, and process parameters to ensure the repeatability and reliability of the experimental results.

To verify the superiority of the proposed model, a dedicated grinding experimental platform has been constructed. The experiments are conducted on a five-axis machining center (DMG Nano60) with a maximum spindle speed of 30000 r/min. The experiments employed SiC ceramic workpieces with dimensions of 50 × 50 × 10 mm, whose material properties are summarized in Table B.1 of Appendix B. The grinding wheel utilized in the experiments is a single-layer electroplated synthetic diamond wheel mounted on a hardened high-speed steel shank, and its detailed specifications are provided in Table B.2 of Appendix B. The grinding forces are measured adopting a three-component dynamometer system consisting of a dynamometer (Kistler 9129A), a multichannel charge amplifier (Kistler 5080A), a data acquisition unit (Kistler 5697A1), and DynoWare software. The normal force and tangential force are continuously recorded at a sampling rate of 20 kHz (corresponding to a sampling interval of 0.05 ms). Fig. 7 shows the experimental setup. During the tests, the workpiece is clamped in a vise mounted on the dynamometer, which is then fixed to the machine table. All grinding experiments are conducted under dry conditions without the utilization of cutting fluid.

To comprehensively cover different grinding conditions, multiple sets of parameter combinations are designed, including a_p, v_f, n_s and M. The experimental parameters are listed in Table 1. In this way, the variations of grinding forces under different machining conditions can

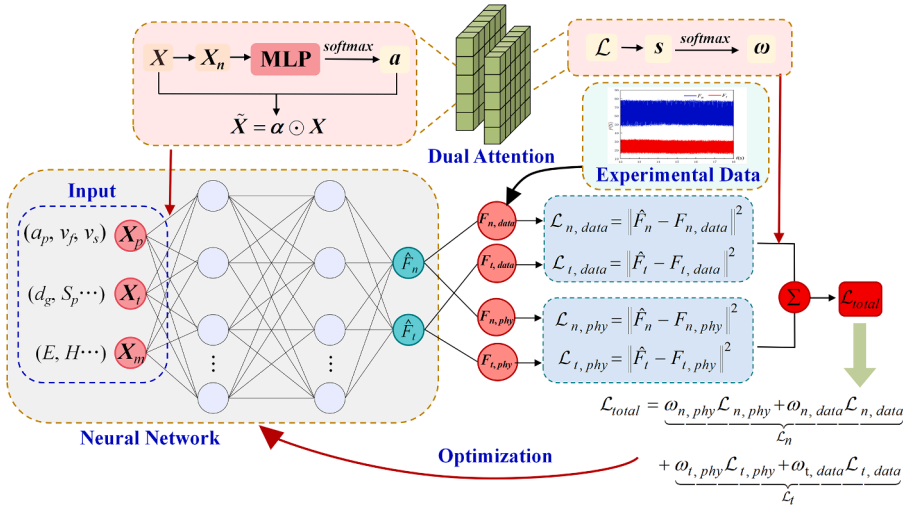


Fig. 6. Overall architecture of the PHGFM. The model takes process parameters, wheel topography parameters, and material parameters as inputs, integrates physical constraints and experimental data through a dual attention mechanism, and optimizes the total loss function by weighting the data-driven and physics-based losses.

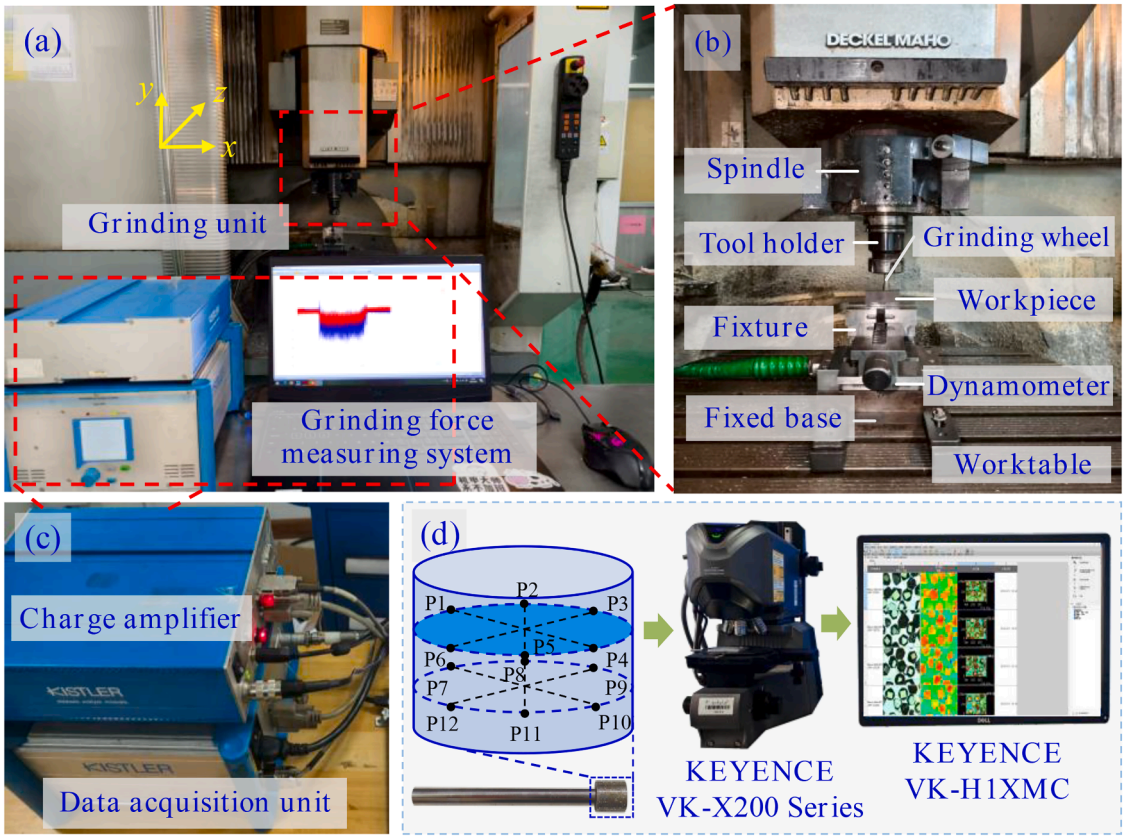


Fig. 7. Experimental platform. (a) Overall experimental setup; (b) Grinding experimental setup; (c) Grinding force signal acquisition system; (d) Grinding wheel surface topography measurement system.

Table 1
Machining parameters in grinding experiments.

Group	Grinding depth a_p (μm)	Feed rate v_f (mm/min)	Spindle speed n_s (r/min)	Wheel mesh size M (#)
1	10, 15, 20, 25	100	7500	100
2	10	50, 100, 150, 200	7500	100
3	10	100	5000, 7500, 10000	100
4	10	100	7500	100, 150, 200, 400

be effectively characterized, thereby providing reliable data support for model validation.

Furthermore, a laser confocal microscope (KEYENCE VK-X200 Series) is utilized to quantitatively characterize the wheel topography. By statistically fitting the measured data and benchmarking them against model predictions, the accuracy of the grinding wheel surface topography model will be further validated (as detailed in Section 3.2).

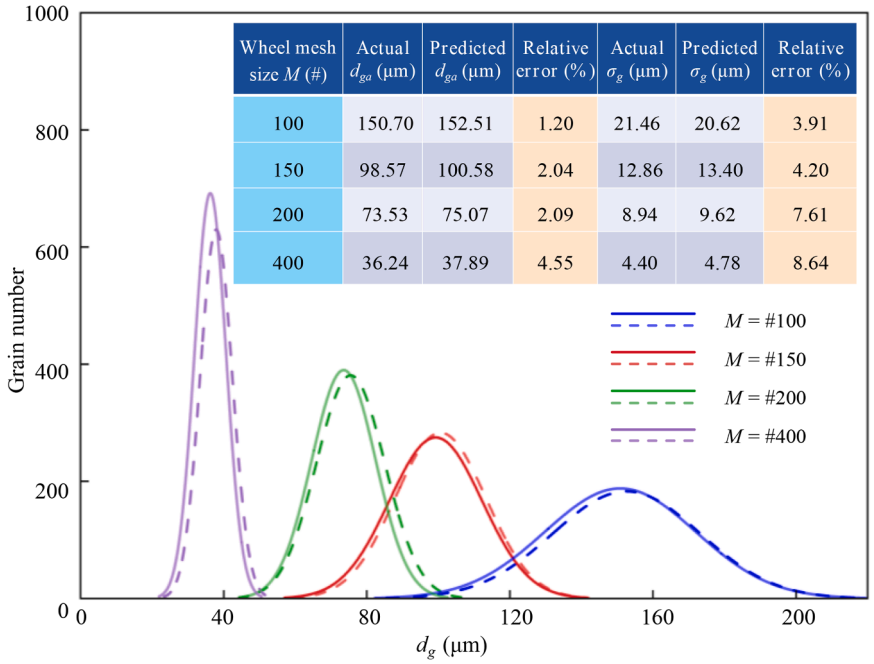
3.2. Experimental validation of grinding wheel topography model

3.2.1. Validation of initial wheel topography model

The wheel topography model forms the foundation of grinding force prediction, and its accuracy directly determines the reliability of the force model. To minimize measurement errors, 12 regions (P1-P12) are uniformly selected, comprising six columns along the circumferential direction and two rows along the axial width, as illustrated in Fig. 7(d). Each measurement region is set to 500 × 500 μm, and five repeated measurements are performed for each region to obtain the average value and ensure accuracy. After acquiring the raw three-dimensional surface, the grain size and spacing are extracted utilizing analysis software. Subsequently, the measurement data are statistically evaluated by constructing histograms and fitting Gaussian distributions.

As illustrated in Fig. 8, the predicted grain size distributions are compared with the experimental measurements for grinding wheels of different mesh sizes. The solid lines represent the experimental measurements, and the dashed lines denote the model predictions. The predicted mean grain sizes show excellent agreement with the experimental values across all mesh sizes, with relative errors below 5 %. Specifically, the deviations for M = #100, #150, and #200 are 1.20 %, 2.04 %, and 2.09 %, respectively, while a slightly larger deviation of 4.55 % is observed for M = #400, which may be attributed to the increased sensitivity of finer grains to measurement uncertainty. The standard deviations σ_g exhibit a similar trend, with errors consistently below 10 %. These results demonstrate that the proposed initial wheel topography model (as defined in Eqs. (1)-(6)) not only captures the mean grain size with high fidelity but also reliably reflects its statistical variability, thereby confirming its accuracy and robustness in characterizing grinding wheels of varying mesh sizes.

Before comparing the modeled and experimentally measured grain spacing distributions, the surface coverage A_s of the grinding wheel needs to be quantitatively evaluated. To achieve this, the original color image is first converted into a single-channel grayscale image. An adaptive thresholding method is then applied to binarize the grayscale image, which enables the separation of abrasive grains from the bond matrix. In the binarized output, the foreground region (abrasive grains) is highlighted in red, while the background region (wheel substrate) is displayed in gray. The ratio of foreground to background pixels is subsequently calculated to determine A_s. The overall processing workflow is illustrated in Fig. 9(a). Utilizing a diamond grinding wheel with 100 mesh grain size as an example, a total of 3394 S_p values are counted,



Wheel mesh size M (#)	Actual d_{ga} (μm)	Predicted d_{ga} (μm)	Relative error (%)	Actual σ_g (μm)	Predicted σ_g (μm)	Relative error (%)
100	150.70	152.51	1.20	21.46	20.62	3.91
150	98.57	100.58	2.04	12.86	13.40	4.20
200	73.53	75.07	2.09	8.94	9.62	7.61
400	36.24	37.89	4.55	4.40	4.78	8.64

Fig. 8. Comparison of predicted and measured grain size distributions for grinding wheels of different mesh sizes. The proposed model achieves a minimum error of 1.2 % in predicting the average abrasive grain size.

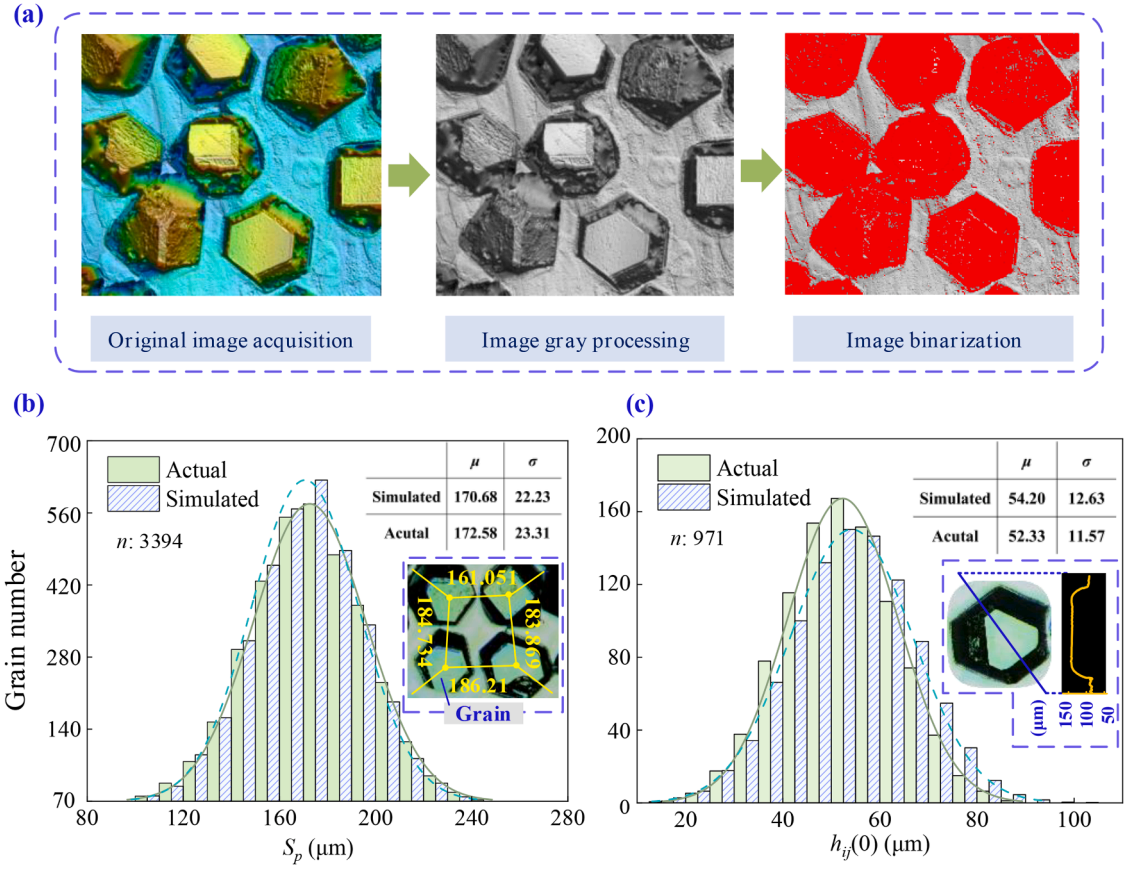


Fig. 9. Characterization of abrasive grain parameters. (a) Workflow of image binarization; (b) Distribution of the grain spacing, with the inset illustrating the measurement process between adjacent grains; (c) Distribution of initial grain protrusion height, with the inset showing the measurement procedure. The grain spacing and protrusion height both follow a normal distribution.

which follow a Gaussian distribution (172.85, 24.41^2), as illustrated in Fig. 9(b). The inset in Fig. 9(b) shows the measurement process for grain spacing. Because direct measurement of abrasive protrusion height is challenging, a cross-sectional method is employed. The inset in Fig. 9(c) displays the shape profile of single grain, where its protrusion height is obtained by calculating the vertical distance from the grain apex to the wheel substrate. The protrusion heights of 971 abrasive grains are measured, which follow a Gaussian distribution (54.20, 12.68^2). The statistical distribution of the protrusion heights is illustrated in Fig. 9(c). The solid lines represent the experimentally measured values, and the dashed lines denote the model predictions. The distribution errors of grain spacing and protrusion height are 4.63 % and 9.16 %, respectively. These results indicate that the proposed initial wheel topography model accurately captures the actual statistical characteristics of abrasive grain topography, thereby validating the reliability.

3.2.2. *Validation of wheel topography reconstruction model*

To validate the wheel topography reconstruction model, several segmented grinding tests are conducted. Under constant process parameters ($a_p = 10$ μm, $v_f = 100$ mm/min, $n_s = 7500$ r/min, $M = \#100$), SiC ceramic workpieces are ground. According to Xi et al. [90], the wheel wear rate progressively decreases as grinding proceeds, provided that no surface burn occurs. At cumulative grinding times of 0.5, 1, 2, 5, and 10 min, the process is intermittently halted, corresponding to material removal volumes (MRVs) of 5, 10, 20, 50, 100, and 300 mm^3, respectively. At each grinding halt, regions P1-P12 are scanned to obtain the protrusion height distribution of the worn grains. Fig. 10(a) presents the representative scanned topographies of the grinding wheel surface at MRV values of 10, 100, and 300 mm^3. Fig. 10(b) illustrates the corresponding distributions of protrusion heights for total MRVs of 10, 50, and 100 mm^3, respectively, where n denotes the number of abrasive grains included in the statistical analysis. To quantify the decay trend, the first four datasets are used to estimate the attenuation coefficients α and β via least-squares fitting, and the relative error and the root mean square error (RMSE)are evaluated for the total MRV of 300 mm^3.

Fig. 10(c) illustrates the decay of the mean protrusion height (μ) and

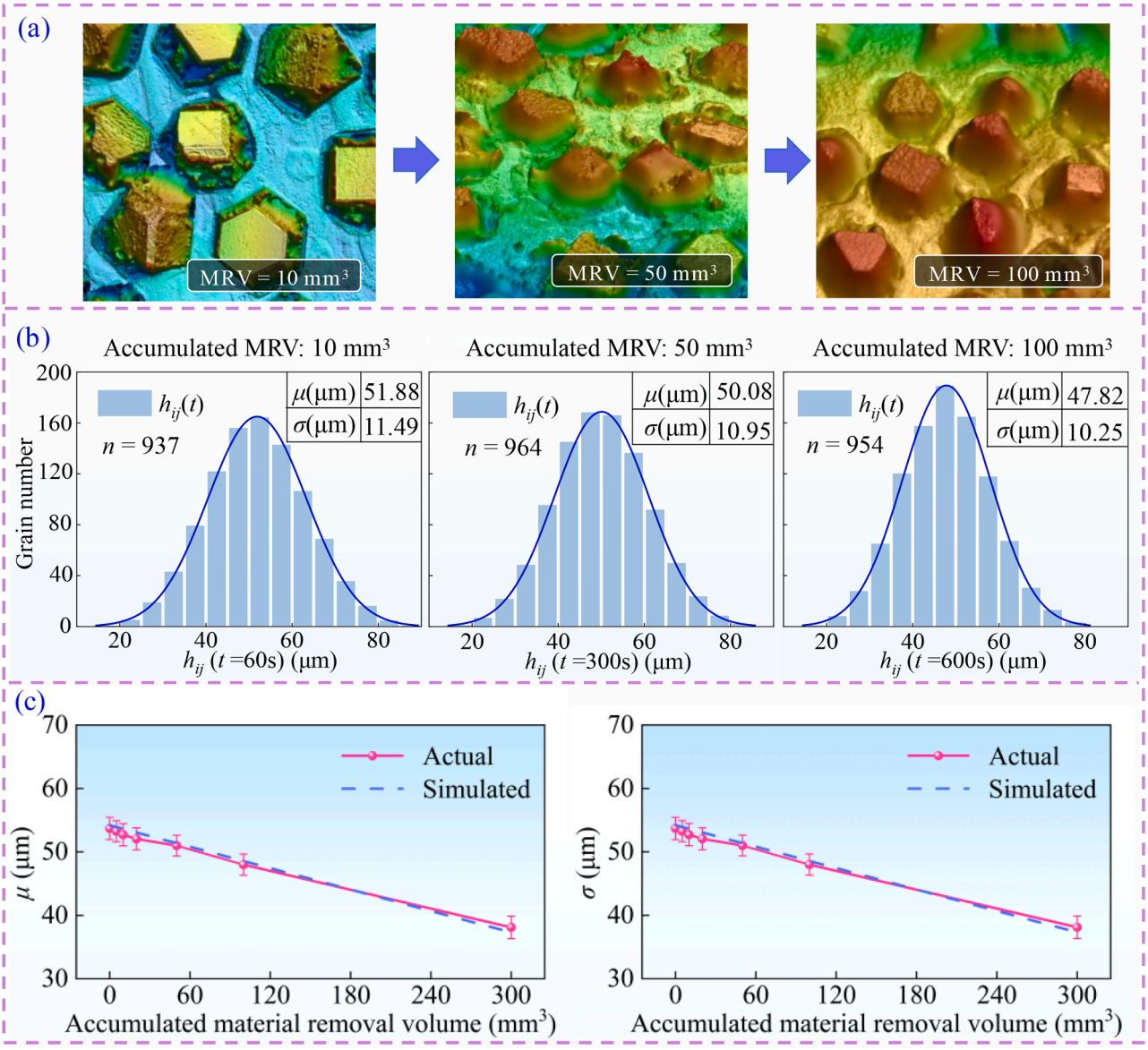


Fig. 10. Validation of the grinding wheel surface reconstruction model. (a) Typical surface topographies of the grinding wheel at different accumulated MRVs; (b) Statistical distributions of grain protrusion heights $h_{ij}(t)$ under different MRVs; (c) Variations in the mean and standard deviations of grain protrusion height with increasing MRV. As MRV increases, the protrusion height of abrasive grains gradually decreases, confirming that the linear decay model can accurately characterize the evolution of the grinding wheel surface morphology during wear.

standard deviation (σ) with increasing MRV. The dashed lines represent the results from three repeated experiments, and the solid curves denote the model predictions. At MRV = 300 mm^3, the relative RMSE is 1.61 μm, with a mean value error of 3.69 % and a standard deviation error of 9.40 %, yielding an overall RMSE of 1.61 μm. Since the experiments are performed over relatively short grinding times under constant process parameters, the statistics of the protrusion height exhibit no segmented features, implying that a linear decay model adequately describes the grain wear behavior under these conditions. Describing the evolution of mean and standard deviation by linear attenuation therefore satisfies the accuracy requirements of grinding force modelling while avoiding the overfitting and added complexity associated with multi-segment parameterization.

3.3. Experimental validation of PHGFM

3.3.1. Model training and interpretability analysis

This section introduces the dataset scale and partitioning strategy. It then analyzes the training stability and convergence behavior through the loss curves and the dynamic evolution of loss weights. Finally, the learned attention weights are visualized, revealing the relative importance of different input features in grinding force prediction.

During the construction of the dataset for model training and evaluation, to eliminate the transitional effects associated with wheel engagement and disengagement, and to ensure data representativeness and comparability across different machining conditions, the central one-third of each complete force signal is retained for model training and evaluation. Following this criterion, 3.03×10^6 effective data points are obtained in total, including 2.20×10^6 data points from experiments varying grinding depth a_p, spindle speed n_s, and wheel grit size M, and 0.833×10^6 data points from experiments with different feed rates v_f.

To objectively assess the model's generalization capability and ensure the rationality of dataset partitioning, a stratified sampling strategy is adopted. Specifically, each individual grinding experiment (i.e., each combination of processing parameters, such as $a_p = 10$ μm, $v_f = 100$ mm/min, $n_s = 7500$ r/min, $M = \#100$) is treated as an experimental unit. During data partitioning, stratified sampling is first performed at the experimental unit level. For example, an experimental unit P1 ($a_p = 10$ μm, $v_f = 100$ mm/min, $n_s = 7500$ r/min, $M = \#100$) generates grinding force signals during the experiment, which are treated as an indivisible experimental unit and assigned in their entirety to either the training set, validation set, or test set. All experimental units are then stratified and allocated to the training, validation, and test sets in proportions of 79 %, 11 %, and 10 %, respectively. Subsequently, all valid data points corresponding to each experimental unit are aggregated into the respective dataset. As a result, the final training, validation, and test sets contain approximately 2.40×10^6, 0.33×10^6, and 0.3×10^6 data points, respectively. Consequently, this partitioning strategy ensures that the grinding force signals obtained under the same process parameter combination are treated as an indivisible whole and assigned exclusively to either the training set, validation set, or test set. As a result, the process parameter combinations included in the test set are never exposed to the model during training. This partitioning strategy enables a realistic evaluation of the model's predictive accuracy under previously unseen process designs, thereby providing a rigorous assessment of its generalization capability.

To analyze the training stability and convergence characteristics of the proposed PHGFM, Fig. 11(a) illustrates the evolution of the data-driven loss, the physics constrained loss, and the total loss as the training epochs increase. As training proceeds, the rates of loss reductions gradually slow down and eventually stabilize, with all losses remaining at low levels thereafter. No pronounced oscillations or divergence are observed, demonstrating the favorable convergence behavior and robust training stability of the model. This visualization provides an intuitive understanding of the evolution of different loss terms throughout the training process, thereby enhancing the interpretability of the model training procedure.

Furthermore, to analyze the dynamic balance between data-driven learning and physical-based constraint, Fig. 11(b) illustrates the adjustment process of the loss weights during model optimization based on the training samples. Specifically, ω_{data} denotes the aggregated weight of the data-driven loss term, and ω_{phy} represents the aggregated weight of the physics-based loss term. It can be observed that, in the early stage of training, the data-driven loss weight is relatively high, which aims to facilitate the rapid establishment of the input-output mapping by the network. Meanwhile, the physics-based loss remains actively involved throughout the training process, continuously enforcing physical constraints on the network predictions. As the training progresses, the two types of weights gradually stabilize, indicating that the model achieves a relatively balanced and synergistic relationship between data-driven learning and physical constraints.

To analyze the model's attention to different input features from an interpretability perspective, Fig. 12 illustrates the distribution of weights learned by the feature attention model for each input feature after training. It should be noted that the weights here reflect the relative importance of different input features, whereas the weights discussed previously are used to describe the relative contributions of the physics-based loss term and the data-driven loss term. It can be seen that the

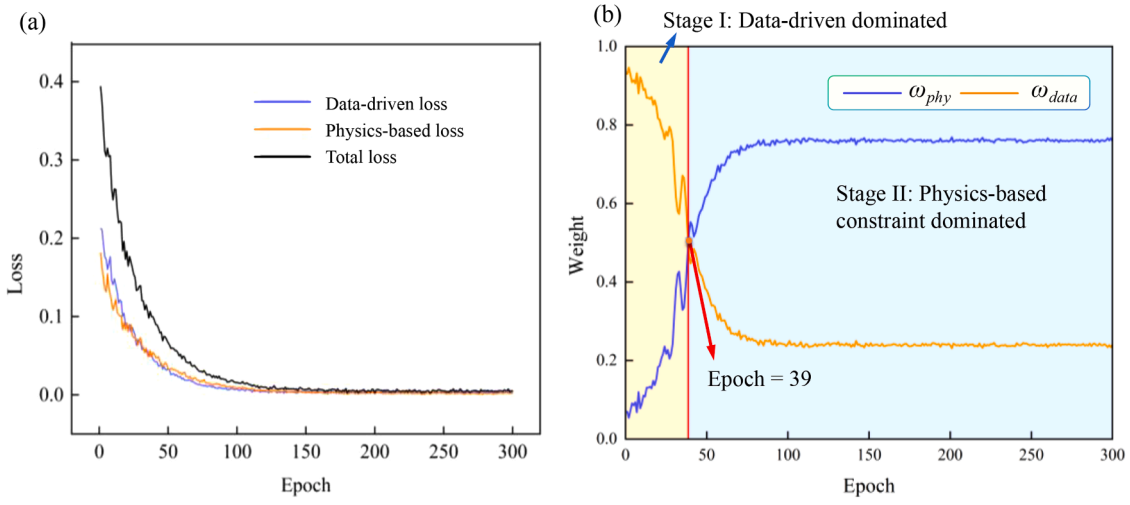


Fig. 11. (a) Loss curves of the PHGFM during training; (b) Evolution of loss weights during training. The loss curves gradually converge and stabilize during training, establishing an effective balance between data-driven learning and physical-based constraints.

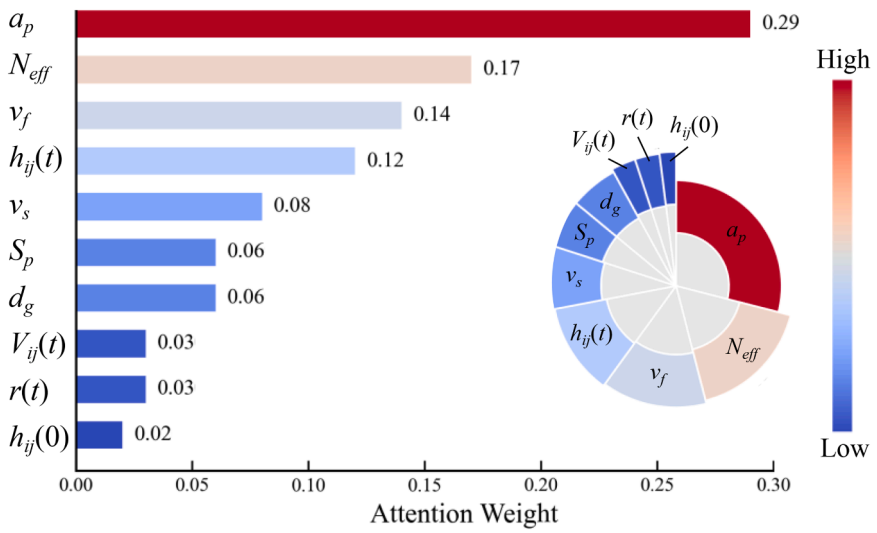


Fig. 12. Distribution of learned feature attention weights. The weights of each input feature after training are illustrated.

feature attention weights exhibit a clear differentiated distribution, in which the grinding depth a_p and the total number of active grains N_{eff} receive the highest attention weights, with values of 0.29 and 0.17, respectively. The model assigns different attention weights to different features, and this result is consistent with the understanding of the grinding force formation mechanism, indicating that the feature attention module can effectively identify and highlight key feature. This mechanism not only improves prediction accuracy but also provides strong support for the physical interpretability of the model.

3.3.2. *Validation of hybrid model under varying machining parameters*

To verify the superiority of the proposed PHGFM in predicting grinding forces, model predictions are compared with the experimental measurements obtained from grinding SiC ceramic workpieces.

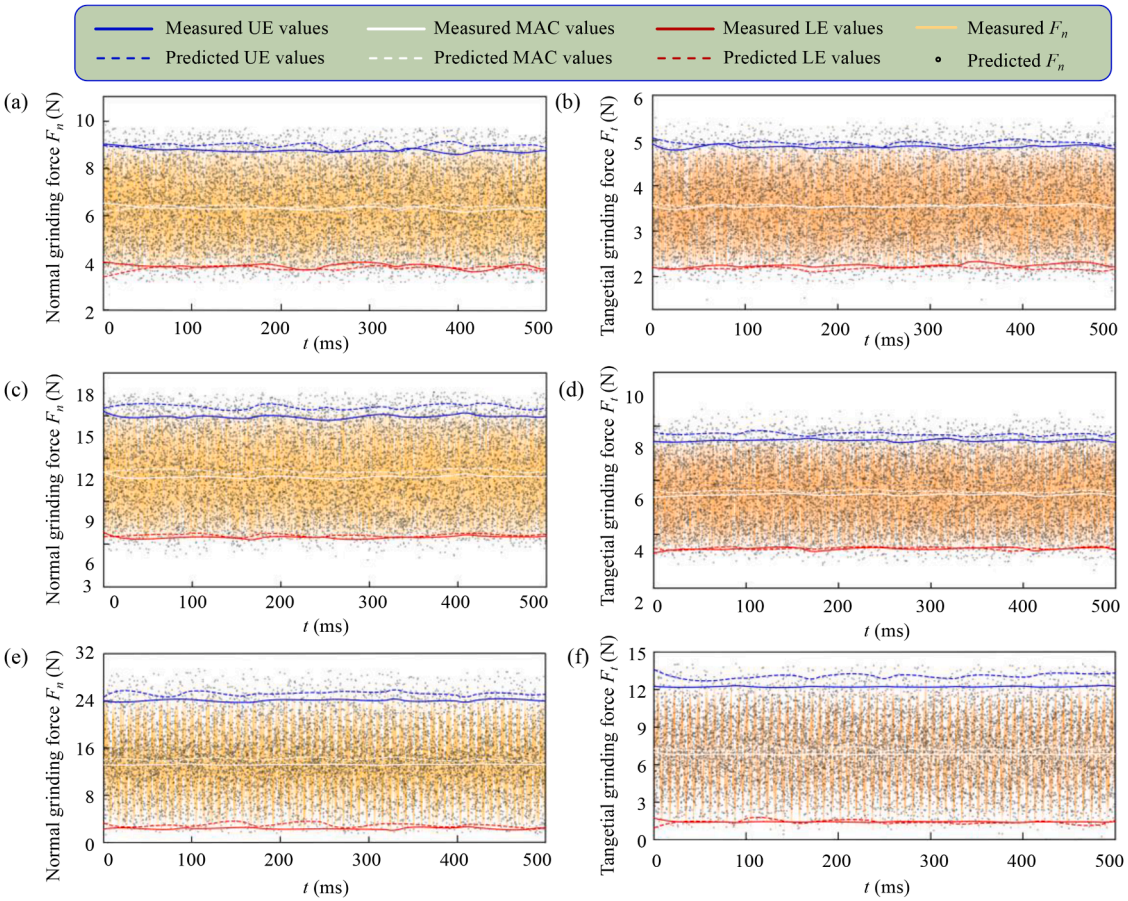


Fig. 13. Comparison of predicted and measured grinding force signals under different grinding depths a_p with constant feed rate ($v_f = 100$ mm/min) and spindle speed ($n_s = 7500$ r/min). Distributions of the normal grinding force for (a) $a_p = 10$ μm, (c) $a_p = 20$ μm, and (e) $a_p = 25$ μm. Distributions of the tangential grinding force for (b) $a_p = 10$ μm, (d) $a_p = 20$ μm, and (f) $a_p = 25$ μm. The grinding force increases continuously as the grinding depth increases.

Since the raw force signals contain transient spikes generated by single abrasive cutting, direct utilization of the maximum or minimum values is highly sensitive to random noise and cannot reliably reflect the overall stable characteristics. Therefore, the raw signals are processed in two steps: (i) a moving window method [76] is applied to smooth the signal and derive the time evolving average force curve (MAC); (ii) within each time window, the maximum and minimum values are extracted to construct the upper and lower envelopes (UE and LE), which delineate the fluctuation bounds of the force signal.

To ensure the reliability of experimental validation, the controlled variable approach is utilized to assess the effects of different grinding conditions. Fig. 13 illustrates the comparison between the predicted and experimentally measured grinding force signals under the constant feed rate of $v_f = 100$ mm/min and the spindle speed of $n_s = 7500$ r/min, while varying the grinding depth a_p (10 μm, 20 μm, and 25 μm). It can be observed that the UE, LE, and MAC values of the experimental and predicted forces are closely matched, demonstrating the feasibility and accuracy of the proposed modeling approach. Beyond visual agreement, a detailed quantitative comparison is provided in Table 2, where UE_{avg}, LE_{avg}, and MAC_{avg} represent the time-averaged values of UE, LE, and MAC, respectively. Among the 18 comparisons, only four exhibited errors exceeding 8 %, while the mean errors of MAC_{avg}, UE_{avg}, and LE_{avg} across all cases are 5.80 %, 5.91 %, and 8.40 %, respectively. Moreover, the minimum mean error is only 3.84 %, which further confirms the high accuracy of the model.

To further evaluate the performance of the proposed model under various machining parameters, Fig. 14(a) and (b) illustrate the comparison results with different feed rates v_f. The error bars denote the standard deviation obtained from three repeated experiments, reflecting the variability and stability of the measured grinding forces under the same machining parameters. When varying v_f, the mean prediction errors for the normal and tangential forces are 7.32 % and 8.33 %, respectively. In addition, the result demonstrates the influence of the feed rate on the grinding forces, revealing an increasing trend with higher v_f. When v_f rises from 100 mm/min to 200 mm/min leads to increments of 3.15 N and 1.66 N in the normal and tangential force, respectively.

Fig. 14(c) and (d) illustrate the comparison results with different spindle speeds n_s. When varying n_s, the average prediction errors of the normal and tangential forces are 6.65 % and 7.39 %, respectively. In addition, as the spindle speed increases from 5000 r/min to 10000 r/min, the grinding forces decrease, with the normal force decreasing by 2.83 N and the tangential force by 1.70 N. Across all tested machining parameters, the proposed PHGFM achieves average prediction errors of 6.76 % for the normal grinding force and 7.28 % for the grinding tangential force, with a minimum error of 3.35 %. These results verify that the model maintains the high accuracy under various grinding conditions.

Table 2
Detailed comparison between experimental and predicted grinding forces: average MAC, UE, and LE values under different grinding depths.

a_p (μm)		Grinding force	MAC_{avg} (N)	UE_{avg} (N)	LE_{avg} (N)
10	F_n	Measured	6.30	9.26	3.33
		Predicted	6.76	9.57	3.11
		Relative Error (%)	7.30	3.35	6.61
	F_t	Measured	3.54	5.15	1.93
		Predicted	3.76	5.46	1.82
		Relative Error (%)	6.21	6.02	5.70
20	F_n	Measured	10.82	15.25	6.16
		Predicted	10.40	16.07	5.84
		Relative Error (%)	3.88	5.38	5.19
	F_t	Measured	5.47	7.51	3.23
		Predicted	5.86	8.09	3.02
		Relative Error (%)	7.13	7.72	6.50
25	F_n	Measured	13.19	24.02	2.37
		Predicted	14.33	25.66	2.82
		Relative Error (%)	8.64	6.83	18.98
	F_t	Measured	6.83	12.23	1.43
		Predicted	7.33	13.38	1.53
		Relative Error (%)	7.32	9.40	6.99

3.3.3. Ablation study

This section conducts an ablation study to systematically evaluate the contribution of each key module in the proposed PHGFM. By selectively removing individual modules and comparing the resulting performance, the effectiveness and necessity of each component within the hybrid framework are quantitatively assessed.

To validate the contributions of the key constituent modules of the proposed PHGFM to grinding force prediction accuracy, a systematic ablation study is conducted. All ablation models are trained and evaluated using the same experimental dataset, ensuring a fair comparison. The performance improvement of the PHGFM primarily benefits from the wheel topography reconstruction module (TR), the UCT refinement (UCTR), physics-based consistent term (PC), and the dual-attention mechanism (DAM). To evaluate the impact of these modules on predictive performance, four comparative ablation models are designed, as detailed below:

Case 1 (w/o TR): The wheel topography reconstruction associated with the wear evolution is removed, and only a static wheel topography is adopted for force calculation, without considering the wear progression during the grinding process.
Case 2 (w/o UCTR): The proposed UCT refinement method is replaced by the classical UCT model [33].
Case 3 (w/o PC): The physics-based constraint term in the loss function are removed, causing the model to degenerate into a purely data-driven deep neural network.
Case 4 (w/o DAM): The dual-attention mechanism is removed, and fixed proportional weighting is applied to both the input features and the hybrid loss components.

Here, w/o denotes without. The prediction accuracy is evaluated using the mean absolute percentage error (MAPE), and the average relative errors of both normal and tangential grinding forces are calculated over all testing conditions. The module configurations and corresponding prediction errors of the ablation models are summarized in Table 3.

The results indicate that the PHGFM achieves the lowest errors in both normal and tangential force predictions with MAPE of 6.278 % and 7.317 %, respectively, demonstrating its superior overall predictive performance. As the key modules are removed, the prediction errors increase to varying extents, indicating that each module contributes positively to the prediction accuracy. Among the ablation cases, Case 3 (w/o PC) exhibits the most pronounced performance degradation, with the relative errors of the normal and tangential forces increasing by 2.996 % and 2.636 %, respectively. This clearly verifies the critical role of the physics-based constraint in improving prediction accuracy and enhancing the generalization capability of the purely data-driven models.

To present the contributions of each module from multiple perspectives, Fig. 15 compares the PHGFM with its ablation variants using RMSE and the R^2 as evaluation metrics. The ablation study comparison results are summarized in Appendix C (Table C.1). The results indicate that the complete PHGFM consistently achieves the best performance on both metrics, exhibiting not only the lowest RMSE but also the highest correlation with the experimental data. The model performance ranks from best to worst as follows: PHGFM > Case 2 > Case 4 > Case 1 > Case 3. In contrast, the RMSE of Case 3 for the normal grinding force increases markedly to 1.036 N, while its R^2 value decreases significantly to 0.924, indicating that the absence of physics-based constraint severely weakens the model's ability to predict the variation patterns of grinding forces.

Therefore, the high-accuracy predictive performance of the PHGFM does not rely on a single module but instead arises from the synergistic

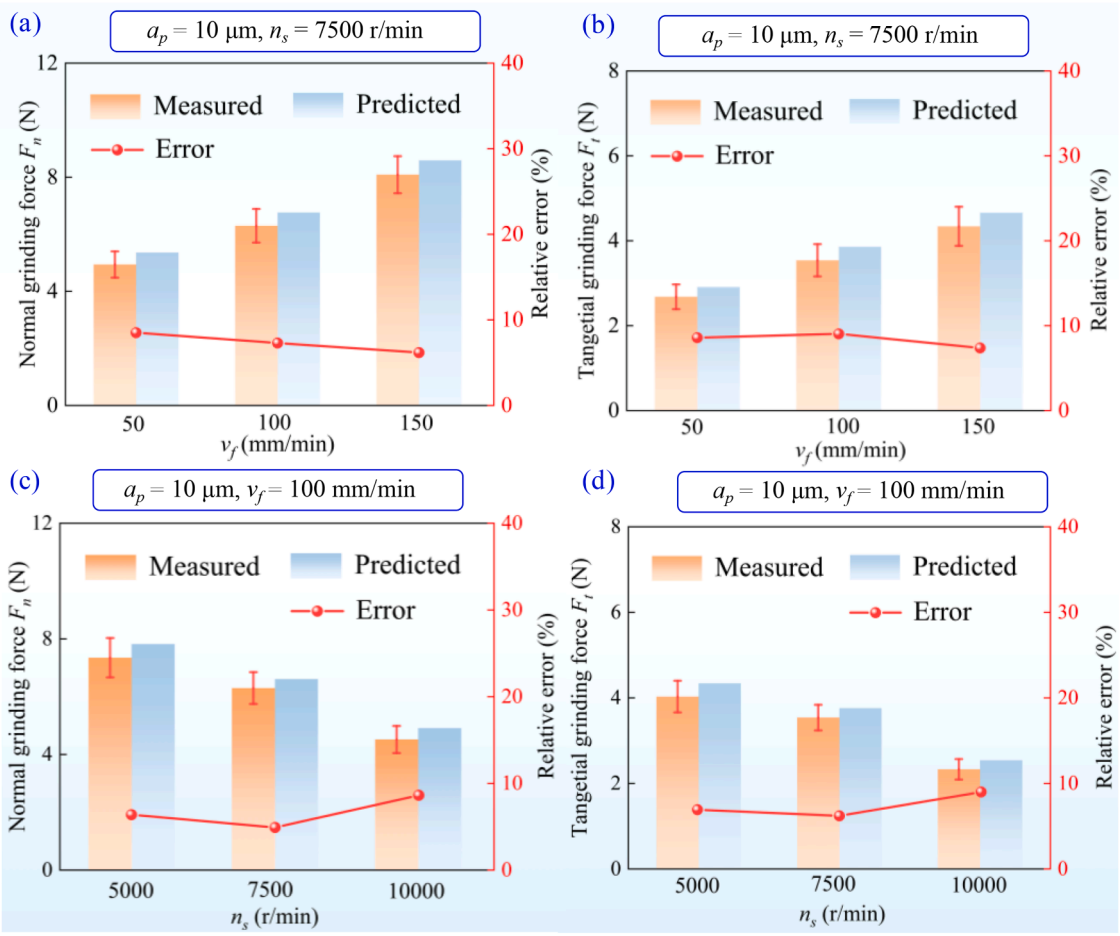


Fig. 14. Comparison between predicted and measured normal and tangential grinding forces under varying v_f and n_s. (a) and (b) illustrate the tangential and normal forces under varying v_f, respectively. (c) and (d) present the tangential and normal forces under varying n_s, respectively.

Table 3
Ablation study results of PHGFM under different module configurations.

No.	Included Modules				F_n		F_t	
	TR	UCTR	PC	DAM	MAPE (%)	Δ (%)	MAPE (%)	Δ (%)
PHGFM	●	●	●	●	**6.728**	-	**7.317**	-
Case1		●	●	●	8.486	1.758	8.943	1.626
Case2	●		●	●	7.754	1.026	8.064	0.747
Case3	●	●		●	9.724	2.996	9.953	2.636
Case4	●	●	●		8.638	1.910	9.018	1.701

#● indicates that the corresponding module is included in the model; Δ represents the error increase relative to the PHGFM baseline, calculated as $\Delta = \mathrm{MAPE}_{\mathrm{Case\ I}} - \mathrm{MAPE}_{\mathrm{PHGFM}}$. All percentage values are calculated based on the entire testing dataset.

integration of all functional components. The incorporation of each module leads to an improvement in prediction performance, which fully validates their complementarity and necessity within the overall framework and further reinforces the rationality of the PHGFM architecture.

3.3.4. *Model comparison and analysis*

To validate the predictive accuracy of PHGFM, we first compare the analytical physical model (Eqs. (1)-(37)) developed in this study with the hybrid model PHGFM. The results indicate that, although the physical model reflects the overall evolutionary trends of grinding forces, its predictive accuracy remains limited. In particular, its representation of the wheel surface topography is constrained, which makes it insufficient to describe the dynamic uncertainties induced by wear evolution, and thus it fails to accurately capture the resulting force fluctuations. By contrast, the hybrid model integrates grinding mechanisms with data-driven learning and a dual-attention mechanism, thereby effectively compensating for the shortcomings of the analytical physical model in handling complex nonlinear mappings. Fig. 16 illustrates the accuracy comparison between the analytical physical model and PHGFM, where Error 1 denotes the prediction error of the improved analytical physical model, and Error 2 corresponds to that of PHGFM. The results demonstrate that PHGFM achieves a 29.16 % reduction in average prediction error compared with the analytical physical model, showing significant improvements across all grinding parameters. These findings confirm that embedding physical constraints into the neural network framework enables an effective balance between physical interpretability and predictive accuracy.

To further evaluate the advantages of the proposed PHGFM compared to purely data-driven methods, this study performs

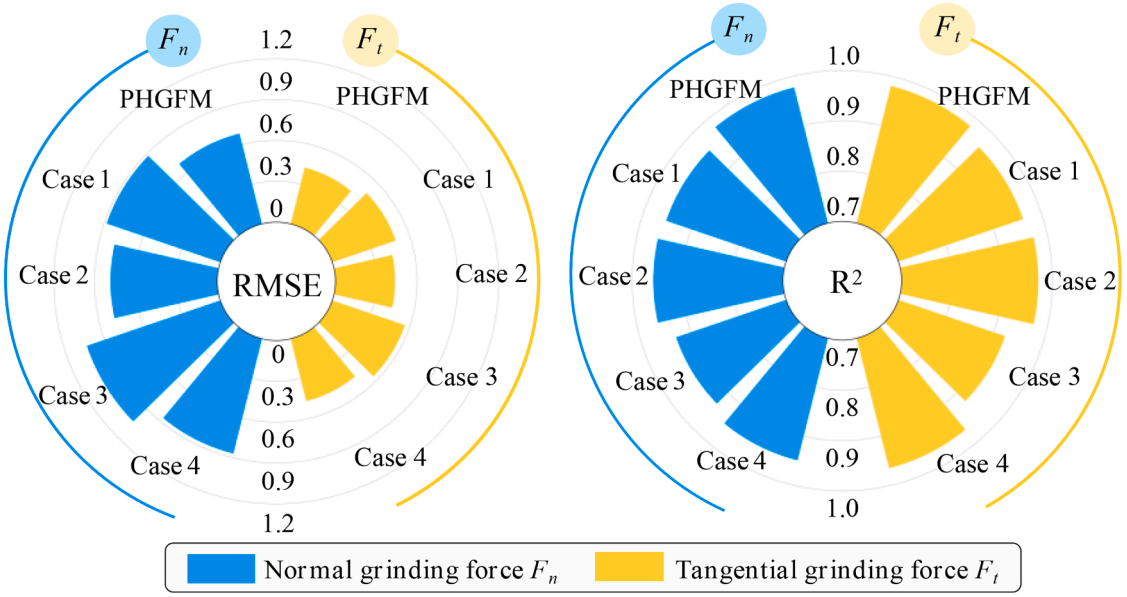


Fig. 15. Influence of different functional modules on the prediction accuracy of grinding forces. Cases 1-4 correspond to the ablation models obtained by removing wheel topography reconstruction (TR), the UCT refinement (UCTR), the physics-based constraint (PC), and the dual-attention mechanism (DAM), respectively.

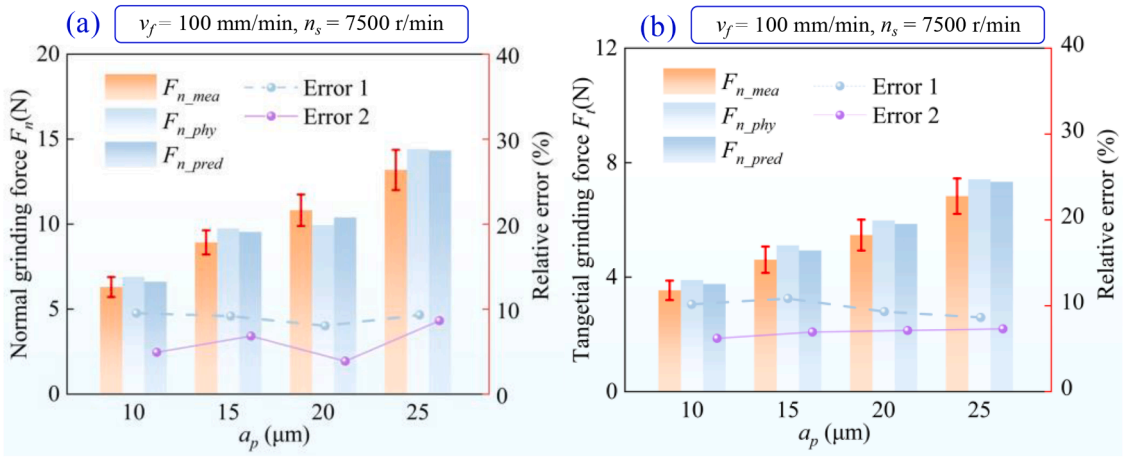


Fig. 16. Accuracy comparison between the pure physical analytical model and the proposed PHGFM. (a) Normal grinding force; (b) Tangential grinding force. The results show that PHGFM achieves significantly higher prediction accuracy, reducing the average error by 29.16 % compared with the analytical physical model.

quantitative comparisons with two representative black-box models: the backpropagation (BP) neural network [57] and the Gaussian process regression (GPR) model [64]. To ensure fairness and comparability, all models, including BP, GPR and PHGFM, are trained and evaluated using exactly the same experimental dataset. The comparative results can be summarized in two aspects.

Specifically, compared to the purely data-driven BP and GPR models, PHGFM achieves significantly lower prediction errors for both the normal grinding force (F_n) and the tangential grinding force (F_t). Fig. 17 further illustrates the performance improvement achieved by PHGFM over BP and GPR. For the prediction of the normal force, the average RMSE and MAPE of PHGFM are 0.681 N and 6.728 %, respectively. Compared to BP and GPR, the average RMSE of PHGFM is reduced by 38.18 % and 46.03 %, respectively, while the corresponding average MAPE is reduced by 36.99 % and 44.98 %. For the tangential force prediction, the average RMSE and MAPE of PHGFM are 0.423 N and 7.317 %, respectively. Compared to BP and GPR, the average RMSE of the PHGFM is reduced by 37.89 % and 47.91 %, respectively, and the corresponding average MAPE is reduced by 25.19 % and 38.19 %.

On the other hand, purely data-driven methods (BP and GPR), due to the lack of guidance from physical prior knowledge, are more prone to overfitting under limited sample conditions. As illustrated in Fig. 17, their correlation coefficients (R^2) of the purely data-driven methods are all below 0.921, indicating the potential instability of the model predictions and limited generalization capability. In contrast, the PHGFM, by embedding the physical mechanisms of grinding force as constraints, maintains high stability under different operating conditions. The R^2 value of PHGFM reaches 0.981, demonstrating its robust performance and excellent generalization ability when faced with unseen conditions.

To provide a more comprehensive comparison and demonstrate the advantages of PHGFM, it is benchmarked against previously reported grinding force prediction models. The selected reference models include: (1) the advanced ceramic grinding force model proposed by Zhou et al. [91], and (2) the model proposed by Fu et al. [58] for predicting grinding forces in Si_3N_4 ceramics. Table 4 summarizes the original parameter backgrounds of the previously reported grinding force models. These machining parameters reflect the initial application contexts of the reference models and are provided to indicate their

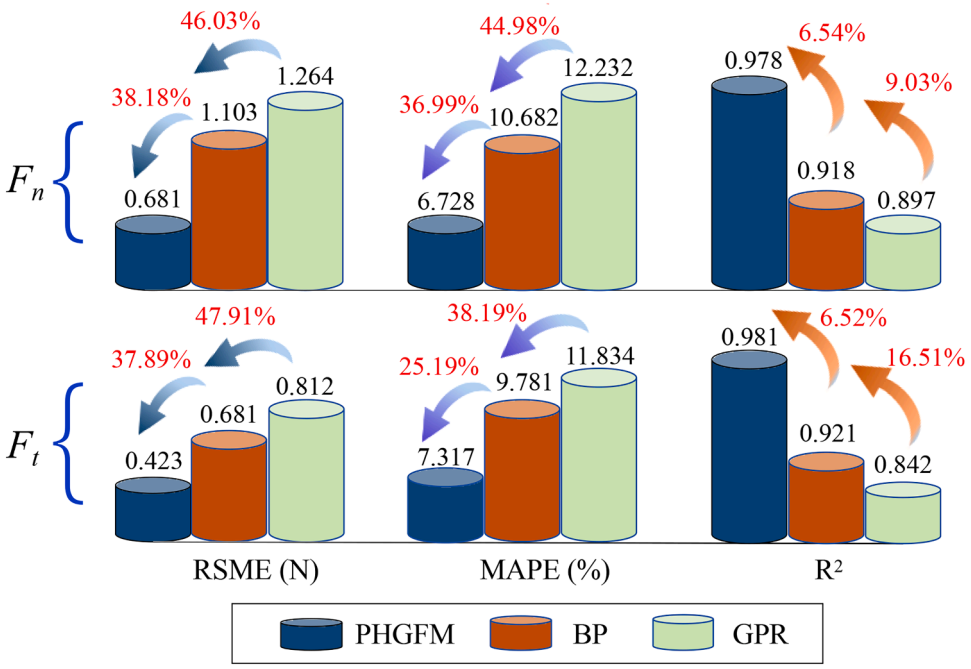


Fig. 17. Improvements of the proposed model compared to BP and GPR models. This shows the improvement of the proposed model in prediction accuracy and generalization ability.

Table 4
Summary of reported grinding force models and associated machining parameters.

No.	Refs.	Workpiece	Wheel	Machining parameters
1	[91]	ZrO_2 ceramic E = 200 GPa, υ = 0.22 K_C= 5 MPa·m$^{1/2}$ H = 12.3 GPa	Mesh: #120 Width: 10 mm Diameter: 15 mm Type: diamond	a_p = 5, 10, 15, 20 μm v_f = 120, 180, 240, 300 mm/min n_s = 2000, 4000, 6000, 8000 r/min
2	[58]	Si_3N_4 ceramic E = 310 GPa, υ = 0.26 K_C = 7 MPa·m$^{1/2}$ H = 16 GPa	Mesh: #120 Width: 10mm Diameter: 10mm Type: diamond	a_p = 2, 5, 10,15, 20 μm v_f = 60, 120, 180, 240, 300 mm/min n_s = 2500, 3500, 4500, 5500 r/min

parameter background and applicable ranges. It should be noted that in this study, all models are re-evaluated using identical machining parameters, material characteristics, and wheel features as inputs, ensuring that the comparison is performed under the same experimental conditions.

The comparison between the experimental measurements of grinding forces and the predictions obtained from different models under identical machining parameters is illustrated in Fig. 18. Specifically, Fig. 18(a) illustrates the comparison of three prediction models with the experimental results at a_p = 10 μm, n_s = 7500 r/min, and v_f = 150 mm/min. Fig. 18(b) shows the comparison of three prediction models with the experimental results at a_p = 10μm, n_s = 5000 r/min, and v_f = 100 mm/min. As observed, Model No. 1 exhibits the largest prediction errors, with average errors of 10.51 % for the normal force and 12.52 % for the tangential force. In contrast, the predictions of PHGFM closely follow the experimental results, demonstrating superior

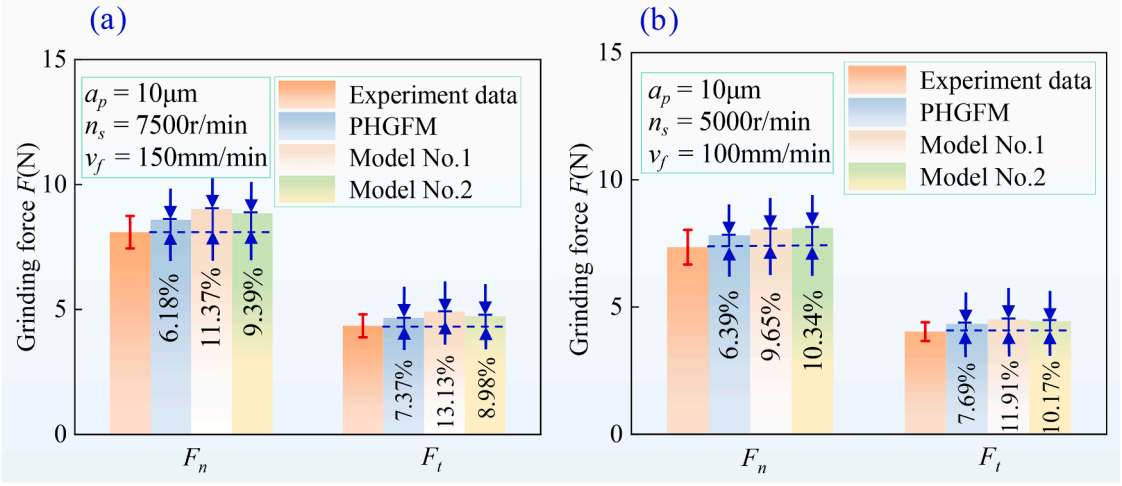


Fig. 18. Comparison of the prediction accuracy of the PHGFM with related studies. PHGFM exhibits the closest agreement with experimental measurements, confirming its superior generalization and robustness across different grinding conditions.

fitting accuracy and stability compared with the other models.

The accuracy of grinding force prediction is critical to process optimization in precision machining. The proposed PHGFM achieves average relative errors of 6.76 % and 7.28 % in predicting the normal and tangential grinding forces, respectively, with absolute deviations typically below 1 N. Under conditions such as larger grinding depths, the relatively increased prediction errors can be attributed to a significant increase in the UCT, which induces nonlinear propagation of lateral cracks and macroscopic brittle fracture. As a result, the actual grinding force deviates from the theoretical value derived under the assumption of linear superposition of multiple abrasive grains.

Overall, PHGFM exhibits excellent predictive capability and robustness across the precision machining regime spanning from ductile-mode grinding to brittle material removal, thereby providing strong support for grinding force prediction under precision grinding conditions.

4. Conclusions

In this study, a physics-informed hybrid grinding force model is proposed to overcome the limitations of existing models in representing grinding mechanisms and improve high prediction accuracy. By coupling the stochastic topography of the grinding wheel with its wear evolution, a topography reconstruction method is established, and an UCT refinement method is further introduced through kinematic analysis. In addition, by embedding the physical mechanisms of grinding force into the PINN framework, the model achieves an integration of physical priors and data-driven learning. The overall framework, combining stochastic wheel topography, UCT refinement, and hybrid neural network modeling, provides a comprehensive approach to achieve precise grinding force predictions.

In terms of wheel surface characterization, a static characterization of the grinding wheel surface is developed, and its topography is reconstructed to account for dynamic wear, achieving a high-fidelity quantification of the wheel's three-dimensional surface evolution during grinding. The initial wheel topography model yields prediction errors of less than 5 % for abrasive grain size and less than10 % for spacing and protrusion height. The wheel topography reconstruction model accurately captures the decay trend of abrasive grain protrusion height, with an average error of 3.69 %. These results lay a reliable basis for the precise calculation of UCT.

Based on the wheel topography model and grain motion analysis, a UCT refinement model is established, which accounts for the wear of abrasive protrusions during grinding, significantly reducing cumulative errors caused by abrasive wear and trajectory interference.

This study further embeds the physical equations of grinding force into the neural network and designs a hybrid loss function that combines physical constraints with data-driven learning. Moreover, a dual attention mechanism is introduced to dynamically balance the influence of physical priors and experimental data, thereby enhancing the model's ability to capture complex nonlinear relationships while maintaining physical interpretability. Overall, PHGFM achieves high prediction accuracy across diverse grinding conditions, with average errors of 6.76 % for the normal grinding force and 7.28 % for the tangential grinding force, and a minimum error of 3.35 %. These results demonstrate the superiority of the PHGFM in capturing the complex mechanisms of grinding force generation. Additionally, this approach provides a reliable tool for optimizing grinding parameters and analyzing grinding stability. Future research will further explore the applicability of the model to a wider range of materials and more complex grinding scenarios.

CRediT authorship contribution statement

Liangdong Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haifeng Ma:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Zhengyi He:** Validation. **Heran Wang:** Visualization. **Jie Chen:** Validation, Software. **Qinghua Song:** Writing – review & editing, Software. **Zhanqiang Liu:** Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Calculation of the critical depths for material removal transitions

As grinding depth increases, the material removal mechanism undergoes a transition from elastic deformation to brittle removal. The critical UCT $h_{e\text{-}d}$ of elastic-to-ductile transition is given by [92]:

$$h_{e-d} = 0.143\frac{H}{E^*}r_a, \tag{A.1}$$

where H denotes the hardness of the workpiece material, r_a represents the tip radius of the abrasive grain, and E^* is the equivalent elastic modulus.

The critical UCT $h_{d\text{-}b}$ of ductile-to-brittle transition can be expressed as [93]:

$$h_{d-b} = 8.7\sqrt{\frac{E}{H}}\left(\frac{K_C}{H}\right)^2, \tag{A.2}$$

where E is the elastic modulus of the workpiece material, and K_C represents the fracture toughness of the workpiece material. The equation considers the relationship between elastic-plastic deformation and crack propagation under microscale contact conditions, providing a theoretical basis for the

quantitative evaluation of the critical depth for brittle material removal [94].

Appendix B. Material and wheel parameters

Table B.1
Material parameters of the workpiece.

Material	Dimensions (mm)	Elastic modulus E (GPa)	Poisson's ratio υ	Hardness H (GPa)	Fracture toughness K_C (MPa·m$^{1/2}$)
SiC	50 × 50 × 10	420	0.16	24.2	4.5

Table B.2
Detailed specifications of the grinding wheel.

Grain type	Diameter D (mm)	Width B (mm)	Mesh (#)	Elastic modulus E_g (GPa)	Poisson's ratio υ_g
Diamond	10	10	100, 150, 200, 400	1050	0.1

Appendix C. Ablation study comparison results

Table C.1
Comparison results of PHGFM and ablation cases.

No.	F_n			F_t		
	RMSE	MAPE (%)	R^2	RMSE	MAPE (%)	R^2
PHGFM	**0.684**	**6.728**	**0.978**	**0.426**	**7.317**	**0.981**
Case1	0.882	8.486	0.952	0.492	8.943	0.962
Case2	0.784	7.754	0.958	0.438	8.064	0.972
Case3	1.036	9.724	0.931	0.563	9.953	0.924
Case4	0.869	8.638	0.950	0.472	9.018	0.966

Data availability

The authors are unable or have chosen not to specify which data has been used.

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Abstract: Accurate prediction of grinding force is crucial for enhancing machining precision. Existing models often rely on simplified assumptions of rigid abrasive grains or mean chip thickness, neglecting the dynamic coupling between wheel surface randomness and grain wear. This paper proposes a physics-informed hybrid grinding force model (PHGFM). By integrating the geometric characteristics and wear behavior of grains, the wheel topography is reconstructed to accurately capture the evolution of the wheel surface topography during grinding. Furthermore, a novel undeformed chip thickness (UCT) model is proposed based on kinematic analysis and dynamic height distribution of grains. Moreover, the physical force computations are embedded into a physics-informed neural network (PINN), where a hybrid loss function incorporating both physical constraints and data-driven components is designed to achieve synergistic optimization of physics priors and data features. The proposed hybrid model demonstrates high predictive accuracy for grinding forces in brittle materials such as SiC ceramics under various grinding conditions. The average prediction errors for the normal and tangential grinding forces are 6.76 % and 7.28 %, respectively, while the minimum errors for the normal and tangential forces reach 3.35 % and 5.70 %, respectively.

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RESEARCH ARTICLE

Design of Blind Guiding Robot Based on Speed Adaptation and Visual Recognition

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ABSTRACT It has become urgent to address the traffic challenges faced by this group with the continuous increase in the number of visually impaired individuals. A blind guiding robot based on speed adaptation and visual recognition was designed to address this problem. The speed adaptation of the robot and the blind person is achieved through feedback control of the distance and speed in this paper. Traffic signals are identified using optimized visual recognition method based on YOLOv5 transfer learning, and man-machine interaction is realized by applying multi-module units such as real-time image, speech, and positioning. The experimental results show that the rate of change of the relative distance was controlled within 13.1%, the relative velocity deviation was controlled within 0.3 m/s, the accuracy of identifying traffic signals reached 91.88%. And when the man-machine distance gap is large, the robot can control the man-machine distance to the set distance within 0.7 s in a timely manner, which effectively ensured the travel safety of blind people and provide the groundwork for the practical application of guiding blind robots.

INDEX TERMS Blind guiding robot, deep learning, Kalman filter, YOLOv5.

I. INTRODUCTION

According to the World Health Organization, at least 2.2 billion people around the world have a vision impairment at present [1]. It is imperative to solve the problem of difficult travel for the blind in this context. Nowadays, blind individuals primarily rely on traditional guide sticks and guide dogs for mobility, with only a few complete smart guide devices available. There is an urgent demand for powerful equipment that can help people obtain a better travel experience and provide them with safer, smarter, and more efficient ways to travel.

Currently, Various types of guide equipment with relatively advanced functionalities for the blind have been developed, including smart guide sticks, wearable guide devices, handheld guide instruments, and mobile guide robots. Chang et al. [2] designed a smart guide stick that combined a variety of sensors and a GPS system for an application based on the Android system, thus realizing real-time transmission of location information. Batliwala et al. [3] developed a guide stick that utilizes a buzzer and vibration motor to communicate information about obstacles ahead to the user timely. Suman et al. [4] designed a guide stick that employed a single-shot detection mechanism and a recurrent neural network to recognize and classify objects in front of the user, which could transmit information about objects to the blind. Dernayka et al. [4], [5] proposed an intelligent guide blind stick equipped with laser detectors and infrared sensors, both of which worked independently. Under low-cost conditions, obstacles can still be detected under strong sunlight.

In addition to guide sticks, other innovative devices have been introduced to assist the blind in different environments. Bai et al. [6] designed a type of guide glass that utilized a dynamic sub-object selection strategy to help the blind avoid dynamic obstacles indoors, thus allowing them to walk safely indoors. Hsieh et al. [7] designed a wearable guide device based on deep learning, thereby realizing the functions of detecting objects and indicating safe routes. Rahman et al. [8] designed wearable guide equipment that employed stereo

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cameras to capture video and applies deep learning algorithms. By integrating this device with web and Android applications, real-time monitoring of the road environment and users can be enabled. Pravin and Sundararajan [9] proposed a handheld guide instrument that used a fixed white LED light as a transmitter and a handheld PIN diode receiver that recorded different frequencies of light to determine the position of the blind.

Inspired by the lane following principle of self-driving cars, Chuang et al. [10] designed a robotic guide dog based on navigation technology and deep convolutional neural network, thus better realizing the function of navigation for the blind. Liu et al. [11] designed a mobile guide robot that employed an improved Dynamic Window Approach (DWA) algorithm to plan the motion path. This reduced the unnecessary steering frequency of the robot, resulting in smoother movements and more timely obstacle avoidance. Du et al. [12] proposed an intelligent blind guiding robot that used the sensor data fusion method based on the D-S evidence theory of the genetic algorithm, which achieved high-precision obstacle recognition and provided essential security for indoor guidance of the blind. Lu et al. [13] proposed an intelligent guide robot that utilized SLAM to estimate the state of the robot and employed the DRL algorithm to predict the best path. It provides navigation information to the users in dynamic environments.

In consideration of the existing shortcomings in blind guidance methods, this study provides a design method of guided robot system for the blind based on speed adaptation and visual recognition. Strive to achieve man-machine speed adaptation and the recognition of traffic signals by integrating technologies such as man-machine interaction, sensors, and audio recognition in this paper, thereby providing more convenience for the blind.

II. THE DESIGN OF A GUIDING ROBOT SYSTEM

This design adopts a self-designed wheeled hexapod robot as the mobile carrying platform, as shown in Figure 1. Combined with a variety of intelligent modules to provide greater independence and autonomy for the blind to travel.

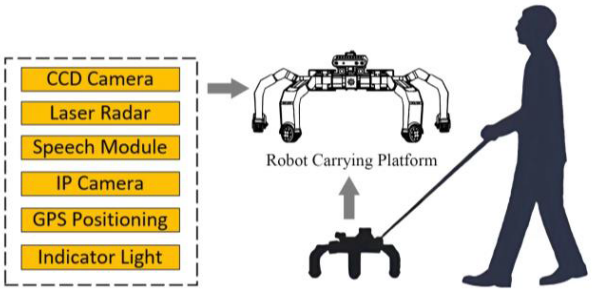

FIGURE 1. System Scheme Composition.

The design of the hexapod robot provides enhanced stability and balance, and can adapt to complex terrain environments. At the same time, the wheel-foot design is added to achieve coordination with the walking speed of the blind under lower cost conditions. The hexapod guiding robot has higher load capacity and better carrying sensors and other equipment. In addition, the hexapod robot can attract the attention of others and provide reliable support for visually impaired individuals because of its unique shape and movement mode.

Inspired by the obstacle avoidance method based on LIDAR by Hutabaratet al. [14], the guiding robot in this design autonomously avoids obstacles by integrating the YDLIDAR X3 Lidar and Hikvision camera. Taking inspiration from Nacir et al. [15] who used YOLOv5 and transfer learning for traffic signal recognition, this study utilizes an optimized YOLOv5 model to recognize traffic signals, thereby improving the safety and independence of visually impaired pedestrians. Referring to the research by An and Lee [16] on robot localization in urban environments, this design combines a GPS module and a network camera to provide positioning and real-time image feedback for blind individuals. At the same time, we add a variety of interactive modules, such as voice control module, indicator light, small speaker, to realize the active control of users and can provide voice and tactile feedback to better convey information to the user. Finally, we also designed a separate control glove for the user, which realized the control of the robot portable.

Therefore, the ''Man-Machine-Environment'' integrated system is summarized, as shown in Figure 2.

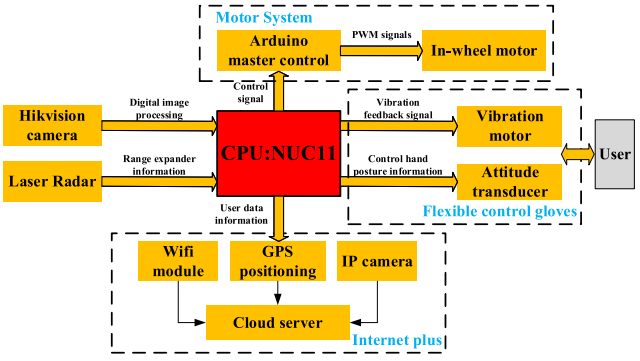

FIGURE 2. "Man-Machine-Environment System" structure composition.

Compared with existing guide robots, this system achieves better man-machine interaction, realizes speed adaptation between the robot and user [17], and can detect and recognize traffic signals.

III. AUTOMATIC SPEED ADAPTATION OF THE MOTOR SYSTEM

Due to variations in walking speed and stride among visually impaired individuals, a fixed guiding speed may be inadequate to meet their needs. Through the implementation of human-robot speed adaptation, a guide robot can automatically adjust its navigation speed according to the user's

walking pace, enhancing comfort and safety during the guidance of visually impaired individuals.

The key technology of the guiding robot is to achieve its speed adaptation to the user. In this design, the YDLIDAR X3 LiDAR mounted on the robot mobile platform was used for distance measurement [18], [19] in order to achieve speed adaptation of the guide robot in response to the user's movements and to track the target. The relative distances x_n and x_{n-1} between the robot and the user were measured at time t_n and time t_{n-1}. By utilizing the uniform velocity model, the relative velocity v_n of the user with respect to the robot at time t_n was calculated as follows:

$$v_n = \frac{x_n - x_{n-1}}{t_n - t_{n-1}} \tag{1}$$

The absolute speed v_{an} of the robot at time t_n was measured by the absolute speed sensor of the robot platform, thus calculating the absolute speed v_{bn} of the user at time t_n:

$$v_{bn} = v_{an} + v_n \tag{2}$$

Owing to systematic and random errors in the range measurement of Lidar, the deviation between the measured value and the real value can lead to a decline in following accuracy and an increase in the possibility of accidents.

To eliminate these errors, this study employed the Kalman filter [20] for error elimination. The Kalman filter [21] is a time-varying optimization linear recursive algorithm primarily known for its observational capabilities. By extracting accurate observations from noisy data, it proves to be advantageous for resource-limited devices.

In this study, a discrete equation was introduced to describe the measurement data of the lidar system.

$$X(k) = AX(k-1) + BU(k) + W(k) \tag{3}$$

In the presence of noise, the system measurements are described as:

$$Z(k) = HX(k) + V(k) \tag{4}$$

In Equation (3) and (4), $U(k)$ represents the system input vector at time k, while $X(k)$ refers to the system state vector at time k. $Z(k)$ denotes the measured value at time k, and H represents the parameter matrix of the measurement system. $W(k)$ and $V(k)$ represent the process and measurement noise, respectively, with covariances Q and R.

By combining the relationship between the predicted value and the measured value and performing relevant parameter updates, the following equation can be derived.

$$X(k|k-1) = AX(k-1|k-1) + BU(k) \tag{5}$$

Equation (5) is the updated Equation of system results, where $X(k|k\text{-}1)$ is the predicted result of the previous state, $X(k\text{-}1|k\text{-}1)$ is the optimal result of the previous state, and $U(k)$ is the currently measured value of the lidar.

$$P(k|k-1) = AP(K-1|K-1)A^T + Q \tag{6}$$

In Equation (6), $P(k|k\text{-}1)$ is the covariance corresponding to $X(k|k\text{-}1)$, $P(k\text{-}1|k\text{-}1)$ is the covariance corresponding to $X(k\text{-}1|k\text{-}1)$, X^{T} is the transpose matrix of A, and Q is the covariance of the system process.

$$X(k|k) = X(k|k-1) + K_g(k)(Z(k) - HX(k|k-1)) \tag{7}$$

$$K_g(k) = \frac{P(k|k-1)H^T}{HP(k|k-1)H^T + R} \tag{8}$$

In Equations (7) and (8), $K_g(k)$ is the Kalman gain, and $X(k|k)$ is the optimal estimated value under the current state, which is the true value of the lidar measurement.

$$P(k|k) = (1 - K_g(k)H)P(k|k-1) \tag{9}$$

In Equation (9), $P(k|k)$represents the covariance of the optimal estimated value that serves as a measure of uncertainty in the estimation process.

Using Equations (5) and (6), the estimation of the lidar returned data is completed, and then the estimation is further processed based on Equations (7) and (8) to refine the results. The Kalman filter algorithm is run recursively on a Next Unit of Computing (NUC11) processor. This enables the estimated distance x_n and speed v_n of the Lidar to gradually converge to their true values.

After the true values of distance x_m and speed v_m are measured, cascade Proportional Integral Derivative (PID) control is adopted. For distance and speed control, a PID controller [22] was used respectively. Equation (10) describes the principle of the PID controller.

$$u(t) = K_p[e(t) + \frac{1}{T_i}\int_0^t e(t)dt + T_d\frac{de(t)}{dt}] \tag{10}$$

First, the measured distance x_m was compared with the set distance x_0, and the deviation $e(t)$, representing the error between them, was obtained. The corresponding coefficients K_p, T_i, and Td were set for the proportional, integral, and derivative controllers, respectively. The outputs of the three controllers were combined to calculate the desired velocity v_e. Then, the expected speed v_e was compared with the measured relative speed v_m. The above process is repeated cyclically to calculate the duty ratio of the Pulse Width Modulation (PWM) signal required for motor control. This controls the operation of the wheel motor, enabling the tracking of a given distance and facilitating man-machine speed adaptation.

In comparison to a single-ring PID control approach, the cascade PID control method makes better use of sensor information, resulting in smoother control, extended robot lifespan, and improved comfort and safety when guiding visually impaired individuals.

IV. THE DETECTION OF TRAFFIC SIGNALS

The detection and recognition of traffic lights is key to guiding robots to guarantee the travel of blind people. This design uses advanced deep learning technology to detect and identify

traffic lights. It has good adaptability and robustness when dealing with complex and changeable environments, and is suitable for real-time monitoring. YOLO (You Only Look Once) is an object recognition and location algorithm based on deep neural network [23] that has a faster running speed and can be used in real-time systems. However, in terms of small target detection, it has a poor effect, and the construction time of the traffic signal data set is long.

This design optimizes the defects of YOLOv5 in detecting small targets and combines the target detection algorithm based on deep learning with the traditional visual recognition algorithm to improve the recognition speed, accuracy, and anti-interference. At the same time, in this study, transfer learning was adopted for model training, which greatly reduced the collection time of data sets.

A. OPTIMIZATION OF YOLOV5

In real traffic conditions, traffic signals may occupy a small number of pixels owing to the large distance or shooting angle. According to the investigation, the down sampling rate of YOLOv5 in convolution is too large, which leads to too few pixels of small targets and fewer target features. Moreover, the first-order SSD algorithm lacks feature fusion, resulting in poor detection for small targets [24].

Based on the target detection algorithm proposed by Google [25], the PANet layer in YOLOv5 was modified to BiFPN (Bi-directional Feature Pyramid Network) in this paper. In addition, this study simplified the backbone feature extraction network, accelerated the feature fusion speed, and effectively reduced the detection time under limited equipment resources. As shown in Figure 3, the red line is the path that transmits semantic information, whereas the green line is the path that transmits location information, thus realizing the two-way integration of top-down and bottom-up deep and shallow features. After fusion, the features are output to the Class and Box networks to detect traffic lights.

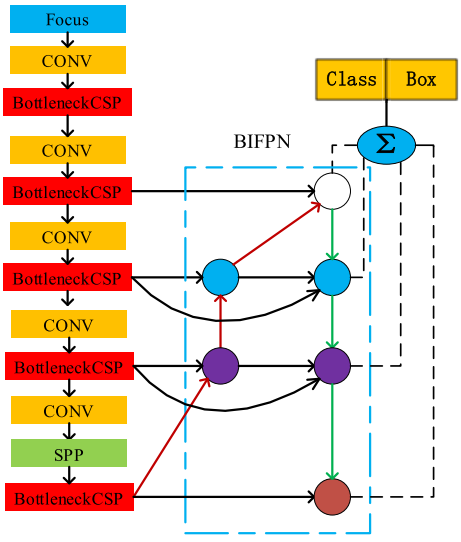

FIGURE 3. YOLOv5 optimization.

B. MODEL TRAINING BASED ON TRANSFER LEARNING

This study uses data sets for model training to train the blind guiding robot. In order to further broaden the data set and diversify the data of the training model, this study independently collected real-time traffic condition pictures from different angles, backgrounds, and lighting conditions as the training set and test set. Part of the image data is as follows:

FIGURE 4. Images of traffic lights.

This study used the transfer learning method. As shown in Figure 5, this study first used labeled data from ImageNet Dataset to conduct preliminary training for the guiding robot. It then uses the homemade data set for secondary training and merges the two data sets to obtain a new data set. Through transfer learning, starting the training model from scratch is avoided, the efficiency of training and learning is greatly improved, the required cost is reduced, and high-quality data sets are obtained. After training the guiding robot several times, it finally realized the recognition of traffic lights.

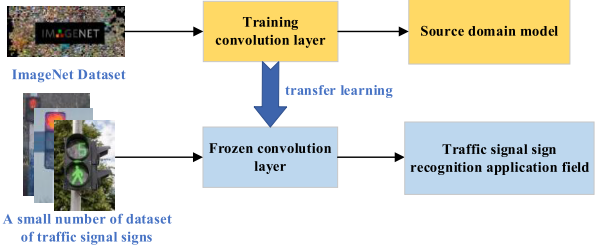

FIGURE 5. Training model of transfer learning.

Finally, the trained model was combined with OpenCV. In the test and real road conditions, the guiding robot can accurately recognize traffic lights, thus providing a safety guarantee for the blind to travel.

V. RESULTS AND ANALYSIS

A. OPTIMIZATION OF YOLOV5

To verify the processing effect of Kalman filter and the effect of cascade PID control in speed adaptation, this study selected a common pedestrian section as the test section. The sampling time was set at 50 ms, the distance was set at 1 m between the robot and the user, and the guide test was conducted. After Kalman filter processing, the man-machine relative distance control and speed adaptation data are shown in Figure 6 and 7, respectively.

As can be seen from Figure 6, the maximum fluctuation range of the original measured distance x_n is 0.58 m, and the noise is relatively large. These noisy data adversely affect the stability of the robot's control system, increasing the risk of potential loss of control."

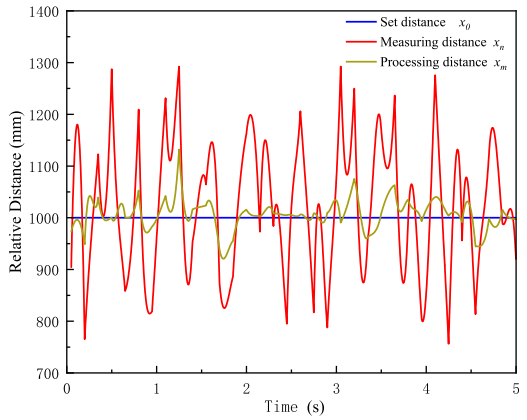

FIGURE 6. Man-machine distance changes with time before and after Kalman filter processing.

After being processed by Kalman filter, the changes in relative man-machine velocity over time are depicted in Figure 7. The average fluctuation range of the original relative velocity v_n is 0.68 m/s. Because the relative velocity v_n is obtained through two different operations of the adjacent distance, the local trend lines of the two groups of data are similar, the overall large fluctuations are obvious, and the noise covariance R is large.

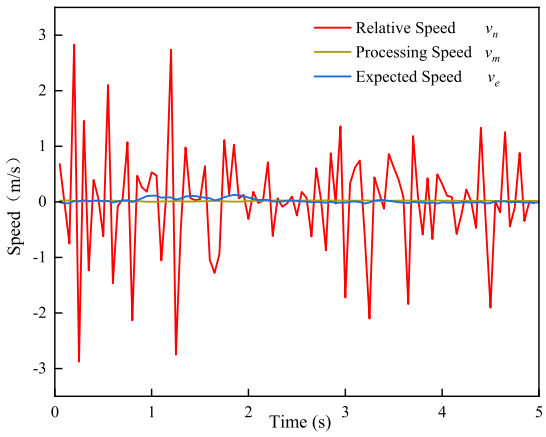

FIGURE 7. Man-machine speed changes with time before and after Kalman filter processing.

From the data processed in Fig. 6 and 7 and Table 1, it is evident that the application of the Kalman filter has significantly improved the results. The average fluctuation range of processing distance x_m drops below 0.06 m, and the average fluctuation range of processing speed v_m drops below 0.19 m/s. The control fluctuated within 13.1% of the set distance x_0, effectively suppressing most of the noise.

TABLE 1. Data characteristics before and after processing.

Data characteristics	Measuring distance x_n(m)	Processing distance x_m(m)	Relative velocity v_n(m/s)	Processing velocity v_m (m/s)
Average fluctuation range	0.17	0.06	0.68	0.19
Standard deviation	0.16	0.04	0.91	0.18

Table 1 shows that after treatment, the standard deviation of relative distance x_m decreases to 0.04 m, and the standard deviation of relative velocity v_m decreases to 0.18 m/s. The speed of the robot controlled by the cascade PID can match the speed of the blind person very well, and the speed of the guiding robot can be truly adaptive.

In practical applications, it is important for the robot to autonomously adapt to the relative distance in order to enter the initial state of the guide mode. When the robot is placed in different initial positions, the autonomous adaptation process of the robot is shown in Figure 8.

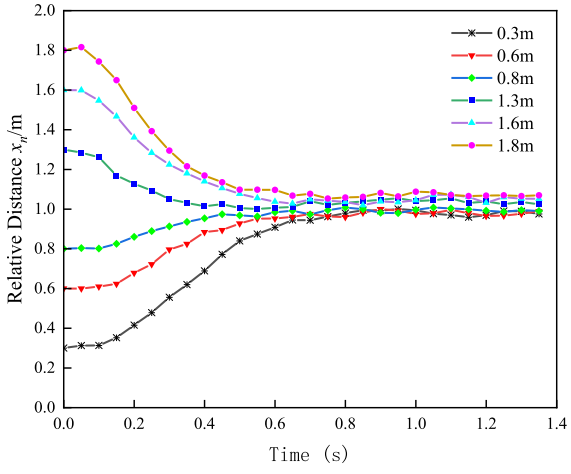

FIGURE 8. Different relative distance autonomous adaptation process.

Under different initial relative distances, the machine can stably control the relative distance to 1 m (set distance x_0) within 0.7 s. Moreover, the average relative velocity value was 1.15 m/s, the speed changed smoothly, and the control was more stable. This study verified the stable dynamic adaptive ability of the guiding robot at different initial distances.

B. TEST OF TRAFFIC LIGHT RECOGNITION

Data augmentation is very important for improving the performance of deep learning [26]. Data collected from traffic light samples as shown in Table 2, then expanded the original data set and fused it with the ImageNet data set to build our own data set.

TABLE 2. Fewer sample data sets.

Model	Total	Proportion (%)
Red	250	43.5
Green	260	45.2
Amber	65	11.3

The guiding robot system was equipped with an Intel NUC11PHi7 microcomputer as the core processor, which adopted the optimized YOLOv5 method to detect and recognize real traffic light images. The results are shown in Figure 9, and the processing time can reach 33 ms. As can be seen from the figure, this method can accurately locate and identify the traffic signals in the image.

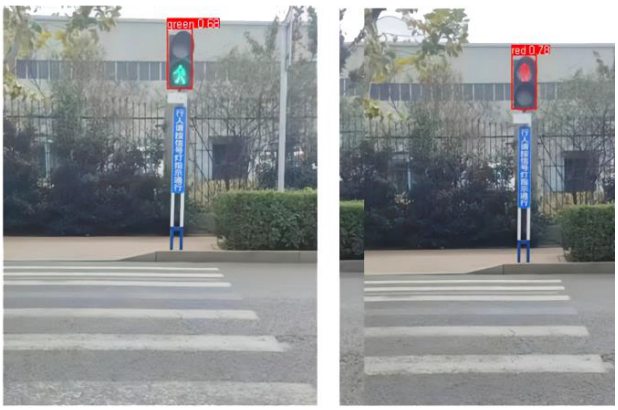

FIGURE 9. Traffic Light Identification Results. (a) green light, (b) red light.

In a real experimental environment, videos of traffic lights in different sections were selected to carry out recognition tests of red, green, and amber lights. Each video contained 120 image frames. The experimental results are listed in Table 3.

TABLE 3. Identification performance of traffic lights.

Target type	Number of Videos	Detection Rate (%)	Recognition Rate (%)
Red	76	97.62	95.43
Green	78	96.95	94.95
Amber	49	95.63	94.32

As can be seen from the experimental results, the detection rate and recognition rate of the guiding robot in a real traffic light environment were more than 94%. The recognition accuracy (detection rate × recognition rate) of each group was 93.16%, 92.05%, and 90.20%, respectively, all exceeding 90%. The overall recognition accuracy rate was 91.88% and the confirmed confidence probability was 0.95. It can be seen that the confidence intervals are (90.58%, 93.16%), indicating that the system can recognize traffic lights accurately, effectively avoid interference, adapt to complex environment, and meet the guiding robot's recognition of traffic lights.

C. ADVANTAGES OF OUR GUIDING ROBOT SYSTEM IN VARIOUS ASPECTS

To demonstrate the advantages of our proposed design for the blind guidance robot system in the current stage of products, we compared it with existing products [27] in terms of obstacle avoidance technology and visual recognition, as shown in Table 3. It can be observed that our robot exhibits outstanding performance in obstacle avoidance technology, visual recognition capability, and user experience.

TABLE 4. Comparative analysis of research.

	Our study	Bai 's wearable guide device	Li 's wearable guide device	Chuang's intelligent guide dog
Obstacle Avoidance Technology	Double guarantee: LIDAR+ CCD camera	Depth camera + ultrasonic range finder	RGB-D camera and IMU real-time detection	Multiple fisheye cameras+ Jetson TX1
The Degree of Visual Recognitio n	Deep Learning (YOLOv5) Accuracy: 91.88%	Identify indoor obstacles	Only identify indoors	Only support identificati on guide line
Contains Functions	Speed adaptation, Traffic signal recognition	Autonomo us path planning	3D acoustic feedback mechanism	High torque motor support handle tension
Sphere of application	Suitable for indoor and outdoor	Suitable for indoor	Suitable for indoor	Only suitable for tracing
Factor of safety	Excellent	Fair	Fair	Good

VI. CONCLUSION

In this paper, a hexapod wheel-foot robot based on independent design is proposed as mobile carrier platform. THE cascade PID control based on Kalman filter processing is used to realize human-machine speed adaptation. And the method of combining the optimized YOLOv5 with transfer learning is used to realize the recognition of traffic lights. The main conclusions are as follows:

(1) The Kalman filter algorithm and cascade PID speed control algorithm are used to achieve autonomous control of the speed of the guiding robot autonomously, and realize the relative distance deviation between the man and machine control within 13.1% and man-machine relative speed deviation control within 0.3 m/s. When the man-machine distance gap is large, the robot can timely control the man-machine distance to the set distance within 0.7 s.

(2) The system realizes the detection and recognition of traffic lights by optimizing YOLOv5 visual recognition algorithm and transfer learning method. The recognition accuracy of each signal lamp is more than 90% and the overall recognition rate reached 91.88% in the real scenario.

In addition, we innovatively apply these methods to the field of guiding robots for individuals with visual impairment, improving the synergy and interaction between guiding robots and users. The incorporation of multiple intelligent modules, such as voice interaction, GPS positioning, and posture control, greatly enhances the capabilities of the guidance robot in assisting individuals with visual impairments. thereby laying a theoretical foundation for the practical application of the guiding robot.

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